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DETAIL SPECIFICATION

ENGINES, DIESEL MARINE, PROPULSION AND AUXILIARY

This specification is approved for use by the Naval Sea System Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers propulsion and auxiliary marine diesel engines.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

HH-P-151 - Packing; Rubber-Sheet, Cloth-Insert

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-3124 - Generator, Alternating Current, 60-Hertz (Naval Shipboard Use)

MIL-DTL-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5

MIL-PRF-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for

MIL-DTL-16884 - Fuel, Naval Distillate

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

MIL-DTL-24696 - Gasket, Sheet, Non-Asbestos

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to commandstandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-167-2	-	Mechanical Vibrations of Shipboard Equipment (Reciprocating Machinery and Propulsion System and Shafting) Types III, IV, and V
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-777	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships
MIL-STD-1474	-	Noise Limits
MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR	-	Protection of Environment

(Copies of this document are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S6430-AE-TED-010	-	Piping Devices and Flexible Hose Assemblies; Volume 1; Technical Directive
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
S9086-CJ-STM-010/075	-	NSTM Chapter 075, Fasteners
S9086-GX-STM-030/220	-	NSTM Chapter 220, Volume 3, Diesel Jacket Water
S9086-HB-STM-010/233	-	NSTM Chapter 233, Diesel Engines
S9086-H7-STM-010/262	-	NSTM Chapter 262, Lubricating Oils, Greases, Specialty Lubricants, and Lubrication Systems
S9407-AB-HBK-010	-	Handbook of Shipboard Electromagnetic Shielding Practices
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@us.navy.mil.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A320/A320M	-	Standard Specification for Alloy-Steel and Stainless Steel Bolting for Low-Temperature Service
ASTM B75/B75M	-	Standard Specification for Seamless Copper Tube
ASTM B98/B98M	-	Standard Specification for Copper-Silicon Alloy Rod, Bar and Shapes
ASTM B111/B111M	-	Standard Specification for Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock
ASTM B135/B135M	-	Standard Specification for Seamless Brass Tube
ASTM B139/B139M	-	Standard Specification for Phosphor Bronze Rod, Bar, and Shapes
ASTM B150/B150M	-	Standard Specification for Aluminum Bronze Rod, Bar, and Shapes
ASTM B169/B169M	-	Standard Specification for Aluminum Bronze Sheet, Strip, and Rolled Bar
ASTM B171/B171M	-	Standard Specification for Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers
ASTM B283/B283M	-	Standard Specification for Copper and Copper-Alloy Die Forgings (Hot-Pressed)
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM D975	-	Standard Specification for Diesel Fuel

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL MARITIME ORGANIZATION (IMO)

IMO MSC/Circular 677	-	Revised Standards for the Design, Testing and Locating of Devices to Prevent the Passage of Flame into Cargo Tanks in Tankers
MARPOL Annex VI and NTC 2008	-	Regulations for the Prevention of Air Pollution from Ships
SOLAS Chapter II-2, Regulation 4	-	High Pressure Oil Fuel Delivery Lines on Small Engines

(Copies of these documents are available online at www.imo.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 8178-1	-	Reciprocating Internal Combustion Engines – Exhaust Emission Measurement – Part 1: Test-Bed Measurement Systems of Gaseous and Particulate Emissions
ISO 8178-3	-	Reciprocating Internal Combustion Engines – Exhaust Emission Measurement – Part 3: Test Procedures for Measurement of Exhaust Gas Smoke Emissions from Compression Ignition Engines Using a Filter Type Smoke Meter
ISO 8178-4	-	Reciprocating Internal Combustion Engines – Exhaust Emission Measurement – Part 4: Steady-State and Transient Test Cycles for Different Engine Applications
ISO 8217	-	Petroleum products – Fuels (Class F) – Specifications of Marine Fuels
ISO 12100	-	Safety of Machinery – General principles for design – Risk assessment and risk reduction
ISO 16852	-	Flame Arresters Performance Requirements, Test Methods, and Limits for Use
ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online at www.iso.org.)

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EUROPEAN SPECIFICATIONS

SIST EN 1127-1 - Explosive Atmospheres – Explosion Prevention and Protection – Part 1. Basic Concepts and Methodology

(Copies of this document are available online at standards.iteh.ai.)

SAE INTERNATIONAL

SAE J1926-1 - Connections for General Use and Fluid Power – Ports and Stud Ends with ASME 1.1 Threads and O-Ring Sealing – Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of this document are available online at www.sae.org.)

INTERNATIONAL ASSOCIATION OF CLASSIFICATION SOCIETIES

UR M66 – Type Testing Procedure for Crankcase Explosion Relief Valves

(Copies of this document are available online at https://iacs.org.uk/resolutions/unified-requirements.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), an engine package shall be subjected to first article test and inspection in accordance with 4.2. This shall include a 1,500-hour endurance test in accordance with 4.2.6.1, unless otherwise approved by the Technical Authority (see 6.3.5). The diesel engine manufacturer shall identify the diesel engine model, proposed horsepower rating, and speed rating, as well as the duty cycle rating for the proposed horsepower and speed. The duty cycle rating shall define the load factor (see 6.6.1), typical operational hours per year, and typical full power operational hours per year. The diesel engine may be procured alone, with accessories, or as a complete package with enclosures (see 6.2).

3.1.1 Engine rating. The engine brake horsepower (HP) rating and speed shall be indicated on the engine identification plate (see 3.10.2). This engine rated HP shall be not greater than the HP value chosen as the 100 percent “rated-load” condition for the endurance test specified in 4.2.6.

3.1.2 Production model. Engines procured under this specification shall be manufacturer’s production models modified to be compatible with environmental conditions, maintenance procedures, and mission requirements peculiar to Navy ship applications. A production model of the engine shall have met at least one of the following requirements:

a. The engine contains no significant changes (see 6.6.6) when compared to the engine that previously passed the first article test (see 4.2) and has been accepted for use by the Technical Authority (see 6.6.8) at the same or a higher power rating and brake mean effective pressure (BMEP). Engines meeting this requirement need to be subjected only to tests required by 4.3.

b. The engine is a current production model and has a background of at least 10,000 hours in two different marine applications comparable to the intended use of the engine being procured. The units considered as verifying this experience factor shall be installed for a rated output at least equal to the rating of the procurement involved. Detailed information shall be provided as to where the units are located, who operates them, and insofar as practical, how they are used. This information will be used to verify the manufacturer’s stated compliance with this requirement. Engines meeting this requirement shall be subjected to the first article inspection specified in 4.2.

c. Manufacturers wishing to introduce an engine not meeting prior testing or experience requirements shall furnish a comprehensive description of the engine design, copies of operating test reports, fuel and lubricating oil consumption, emission levels, and other evidence that would attest to the integrity of the engine. A determination will then be made as to any tests, in addition to those specified in 4.2, which would be required.

3.1.3 Specific fuel consumption. The specific fuel consumption of the proposed diesel engine shall be identified across the diesel engine power band for the standard operating conditions specified in 3.3.1.

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3.1.4 Operability. The diesel engine shall be designed for continuous operation at each point of the specified speed range and power range up to and including rated power while operating at the standard operating conditions specified in 3.3.1. The diesel engine manufacturer shall define the period between major overhauls when operating continuously at rated power. Other requirements for operability are as follows:

a. The diesel engine design shall accommodate transient thermal gradients following emergency shutdowns (no cooldown period) and shall permit immediate restarting.

b. The diesel engine shall be designed to allow for operation without damage up to the overspeed trip speed of the diesel engine.

c. The diesel engine shall be designed for extended shutdown periods of up to 7 days without requiring any special maintenance procedures.

d. The engine shall be designed to allow for operation without damage up to 110 percent HP rating of the engine.

3.2 Materials. Materials used to construct the diesel engine and its accessories shall be the same material used in standard commercial marine diesel engines. Other general requirements for materials are as follows:

a. All materials shall be resistant to, or protected to resist, saltwater corrosion and corrosion from other fluids used in operating, maintaining, or servicing the diesel engine in a marine environment.

b. Only corrosion-resistant materials shall be used in seawater system piping and components. Coated or painted cast iron or other ferrous metals shall not be used.

c. Contact between dissimilar metals shall be avoided in order to minimize galvanic corrosion. An interposing material that will minimize or eliminate this galvanic effect may be used.

3.2.1 Aluminum parts. Aluminum parts other than pistons, blowers, fuel pumps, and turbochargers, shall be electrically isolated from dissimilar metals and protection from corrosion.

3.2.1.1 Threaded holes. Threaded holes in aluminum intended for bolts or screws for components or accessories normally removed or disassembled during routine maintenance or examination shall have steel inserts. Where insufficient basis metal exists, studs may be used without inserts.

3.2.2 Copper parts. Copper shall not be used in fuel or lubricating oil piping, except for Monel-type (Unified Numbering System (UNS) 04400) copper alloys or equal may be used in lubrication oil piping.

3.2.3 Coatings. Paint, plastic, or zinc coatings shall not be applied to the interior of gear cases, lubricating oil sumps, or any other interior surface that will be in contact with fuel, hydraulic, or lubricating oils.

3.2.4 Prohibited materials. The engine, enclosure, sensors, actuators, or other engine related hardware shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.2.5 Health hazard. Material hazards shall be identified, assessed, and mitigated in accordance with MIL-STD-882.

3.2.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Environmental conditions.

3.3.1 Standard operating conditions. The following operating conditions shall be used for the design and testing of the engine and accessories:

a. Ambient air temperatures: -40 to 51.7 °C (-40 to 125 °F) (Enclosed Spaces) -40 to 51.7 °C (-40 to 125 °F) (Open Deck)

b. Barometric air pressure (dry): 635 to 812.8 millimeters of mercury (Hg) (25.8 to 32 inches of Hg)

c. Freshwater from engine: As per engine manufacturer design

d. Maximum exhaust system back pressure: 0 to 127 millimeters of Hg (0 to 5 inches of Hg)

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e. Seawater inlet temperature: 1.7 to 35 °C (35 to 95 °F)

f. Seawater outlet temperature: As per engine manufacturer design limits

g. Lubricating oil temperature from engine: As per engine manufacturer design

h. Fuel temperature to engine: 35 °C (95 °F) (maximum)

i. Maximum intake air system depression using clean filter elements:

(1) Naturally aspirated: 381 millimeters of water (15 inches of water)

(2) Turbocharged/Supercharged: 304.8 millimeters of water (12 inches of water)

3.4 Diesel engine design.

3.4.1 Inclination. The engine, accessory components, and piping systems shall be capable of operating in accordance with all the requirements of this specification and without loss of fluids, loss of lubricating oil pump suction, and without the connecting rod and gears dipping into the oil under the following conditions:

a. When the ship is permanently trimmed down by the bow or stern up to 5 degrees from the normal horizontal plane

b. When the ship is permanently listed up to 10 degrees to either side of vertical

c. When the ship is pitching 10 degrees up or down from the normal horizontal plane

d. When the ship is rolling up to 45 degrees to either side of vertical

e. Unless otherwise specified (see 6.2), the engine installation shall be parallel to the centerline of the ship

3.4.2 Shock resistance. When specified (see 6.2), the propulsion unit including the engine and, if used, marine attached and detached accessories shall meet the grade A requirements of MIL-DTL-901 as modified in 4.2.7.18. Auxiliary engines shall be tested under the shock requirements of the driven equipment.

3.4.3 Engine alignment. The alignment between engine and dynamometer or between engine and generator shall be measured both before and after testing to verify there has been no change after testing.

3.4.4 Compression and firing pressure. If an engine is equipped with a means to measure compression and firing pressures, engines shall be tested in accordance with 4.2.7.10 to verify pressure remains within manufacturer's specification.

3.5 Reliability and maintainability.

3.5.1 Reliability. Reliability will be measured for Navy ship application by analysis of results of the 1,500-hour endurance test conducted in accordance with 4.2.6. An acceptable level of reliability shall be demonstrated when the engine has completed the entire 1,500-hour cyclic test with no forced shutdowns (see 4.2.3.10.3). Repairs or adjustments shall not be made to the engine during this test except as may be specifically approved on a case-by-case basis by the Technical Authority, their designated representative, or as authorized in 4.2.3.10. Engines must be able to be supported shipboard for at least 90 days, assuming constant operation at 80 percent load and 100 percent speed. Spare parts and consumables necessary to support 90 days will be considered by the TWH as part of reliability (see 6.7).

3.5.1.1 Mean time between overhaul (MTBO). Medium-speed diesel engines shall have a 20,000-hour minimum MTBO, and high-speed diesel engines shall have a 10,000 hour minimum MTBO.

3.5.2 Maintainability. When specified (see 6.2), a maintenance engineering analysis (MEA) (see 6.6.2) shall be conducted prior to design testing for each functional system, equipment, unit, assembly, or subassembly that requires maintenance and the continued operation of which is essential to the successful completion of a mission itself.

3.5.2.1 Accessibility. The design shall provide accessibility to all parts that require examination, inspection, preventative maintenance, or replacement in service with the minimum practicable need for disconnection or removal of another part or assembly. Access panels, covers, peep holes, etc. shall be employed to the fullest extent feasible and compatible with the intended examination or maintenance action.

3.5.2.1.1 Engine attached accessories. Engine attached accessories such as pumps, blowers, turbochargers, and intake air coolers shall be installed in such a manner as to permit accessibility to the accessories and the engine.

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Engines shall be provided with openings that have removal covers in the frame and crankcase above the mounting flange for servicing and examining the engine.

3.5.2.1.2 Panel instruments. Panel instruments shall be accessible for reading from the operating station and for maintenance and calibration check.

3.5.2.1.3 Personnel access. Permanent gratings or deck plates shall be installed to allow personnel access for monitoring and maintenance. Engine attached piping and cabling shall be installed in a manner that will not interfere with routine monitoring and maintenance.

3.5.2.1.4 Minimum required space. The envelope of the minimum required space around the unit, which is needed for ease of maintenance, shall be shown in the installation drawing.

3.5.2.2 Interchangeability. All parts, including repair parts furnished under the same contract or order or manufactured to the same drawings, shall be interchangeable without the necessity for further machining, selective assembly, or hand-fitting of any kind to the fullest extent feasible.

3.6 Construction.

3.6.1 Accessory equipment. Unless otherwise specified (see 6.2), the following accessories shall be furnished with each engine:

a. Seawater pump (see 3.7.2.1)
b. Freshwater pump (see 3.7.2.1)
c. Lubricating oil pressure pump (see 3.7.4)
d. Lubricating oil priming pump (see 3.7.4)
e. Fuel supply pump (see 3.7.3)
f. Fuel oil priming pump (see 3.7.3)
g. Lubricating oil cooler (see 3.7.4.2)
h. Freshwater/seawater heat exchanger (see 3.7.2.2.3)
i. Lubricating oil filter(s) (see 3.7.4.3)
j. Lubricating oil strainer (see 3.7.4.4)
k. Intake air silencer (if required) (see 3.7.5.1)
l. Intake air cooler (if turbocharged) (see 3.5.2.1.1)
m. Fuel oil filter (see 3.7.3.3)
n. Fuel oil strainer (see 3.7.3.3)
o. Temperature regulator or thermostat (see 3.7.2.3)
p. Governing system (see 3.7.8)
q. Overspeed trip device or governor (see 3.7.8.4)
r. Starting system (see 3.7.7)
s. Blower (if required for two-cycle engines) (see 3.7.5.4)
t. Engine-mounted controls (see 3.7.10)
u. Lubricating oil low-pressure alarm switch (see 3.7.9)
v. Freshwater high-temperature alarm switch (see 3.7.9)
w. Fuel oil booster pump (if required) (see 3.7.3)
x. Fuel oil cooler (if required) (see 3.7.2.2.3)
y. Cylinder test and indicator cocks (see 3.7.16)
z. Engine turning or jacking device (see 3.7.15)
aa. Remote shutdown system (see 3.7.11)
bb. Turbocharger (if used) (see 3.6.3)

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cc. Instruments and panel (see 3.5.2.1.2)

dd. Emergency shutdown device (see 3.7.11)

ee. Pyrometer (see 3.7.9)

NOTE: Other accessories or engine attached piping necessary to facilitate engine operation, other than those specified herein (see 6.2), shall be furnished. Unless otherwise specified (see 6.2), piping including valve fittings and flanges, alarm systems, wiring, or mechanical linkage required to connect detached accessories, instruments, and controls to the engine, or to cooling water, fuel starting air, electrical, exhaust, or lubrication system will be furnished by the installing activity.

3.6.2 Guards and shields. Guards and shields shall be furnished in accordance with 3.6.2.1 and 3.6.2.2.

3.6.2.1 Guards. Guards for protection of personnel will be provided for flywheels and other exposed moving parts. Where there are no feasible means for attaching the guard to the equipment, the guards will be provided for installation by the installing activity (see 6.2).

3.6.2.2 Shields. Hot machinery surfaces shall be shielded or insulated so that no external surface of the machinery shall exceed a temperature of 220 °C (428 °F). Asbestos insulating products or materials shall not be used. Fuel and lubricating oil filters, strainer, charge air system, aftercooler, lines, or fittings, including gage lines, shall be located or shielded so that any leaks cannot drip or spray on any component with a surface temperature of 220 °C (428 °F) or higher or into the engine intake. Acceptable shielding requirements for different piping systems can be found in S9086-HB-STM-010/233.

3.6.3 Turbochargers. Turbochargers (if used) shall be capable of containing within the housing any parts of the rotating members that may become fragmented due to loosening, shattering, or over-speeding.

3.6.4 Crankcase fumes. A system to dissipate all fumes generated by the engine in the crankcase and other areas, standard with the engine manufacturer's design arrangement, shall be provided.

3.6.5 Engine crankcase and air box explosion protective devices or design. Explosion relief valves of an approved type shall be installed on enclosed crankcases of all engines having a cylinder bore of 200 mm (8 inches) and above or having a crankcase gross volume of 0.6 m^3 (21 ft^3) and above. Relief valve design and construction shall be in accordance with the following accepted industry standards:

a. ISO 16852

b. ISO/IEC 17025

c. SIST EN 1127-1

d. ISO 12100

e. IMO MSC/Circular 677

3.6.5.1 Crankcase explosion relief valves. Crankcase explosion relief valves, produced in accordance with UR M66, are to be provided with a copy of the manufacturer's installation and maintenance manual that is pertinent to the size and type of valve being supplied for installation on a particular engine.

3.6.6 Welding. Welding and allied processes shall be in accordance with S9074-AR-GIB-010/278 unless specific waivers have been or are granted by the Technical Authority.

3.6.7 Threaded fastening. Threaded hardware parts shall conform to S9086-CJ-STM-010/075. Nuts and other threaded fastenings on the engine and accessories shall be secured by wire, lock washers, cotter pins, clips, or other means, in such manner as to positively prevent the part working loose under operating conditions. Drain plugs shall be readily removable with general purpose wrenches. Threaded holes in aluminum intended for bolts or screws for components or accessories normally disassembled during routine maintenance or examination shall have steel coil inserts. Studs may be used without inserts if they do not have to be removed for maintenance.

3.6.8 Timing marks. Means shall be provided to permit checking of valve, injectors, start air injection, and injection pump timing. Means for timing and barring the engine over by hand shall be incorporated in the engine design.

3.6.9 Crankshaft. Nuts on coupling flange bolts shall be accessible. Provisions shall be made on the free end of the crankshaft for attaching a torsiograph drive.

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3.7 Fluid piping systems.

3.7.1 Piping. Engine attached piping shall be strapped and supported by the engine structure to prevent vibration and resist shock. Take-down joints in fuel systems shall be kept to a minimum to reduce leaks. Connections from the engine to the ship's systems (i.e., exhaust, lubricating and fuel oil inlet and return, and the raw water supply and discharge) shall be in accordance with MIL-STD-777. Flange connections shall be either commercial or Navy type. If commercial flanges are supplied, each connection to the ships piping systems shall be provided with a companion flange for welding to the ships piping. Navy flanges for steel or nonferrous piping and fittings shall be in accordance with MIL-PRF-20042. Except as exempt (see 3.7.1.1), taper pipe threads are not permitted in engine piping systems, accessories, or driven equipment. Thread connections, where used, shall employ straight threads in accordance with SAE J1926-1.

3.7.1.1 Pipe threads. Taper pipe threads are permitted in the following areas:

a. Pipe plugs in sizes of 19.1 millimeter (¾ inch) and below used for applications where design gage pressures do not exceed 0.0146 kilogram per square meter kg/m^2 (50 pounds per square inch (lb/in^2)). Plugs shall be seal welded or brazed, if possible.

b. Instrumentation, controls, vent, filling, and drain connections where design gage pressures do not exceed 0.0146 kg/m^2 (50 lb/in^2) and where fluids handled:

(1) are not toxic.

(2) are not dangerous.

(3) would not cause atmospheric contamination.

(4) would not create a fire hazard and, in the event of failure, would not cause a major breakdown of equipment, would not create a hazard to the surrounding area, and would not affect the operation of other vital equipment.

c. Pipe plugs internal to the engine.

3.7.1.2 Flexible hose. If used, flexible hoses shall be in accordance with S6430-AE-TED-010.

3.7.2 Cooling water system. Engines shall operate with a closed freshwater cooling system.

3.7.2.1 Cooling water pumps. Freshwater and seawater pumps shall be of the centrifugal type driven by the drive arrangement capable of preventing water leakage into the engine lubricating oil system. Where practical, the freshwater and seawater pumps shall be interchangeable. Pumps for direct reversing engines shall be capable of operating in either direction. The pumps shall be so mounted or provided with a continuous vent to prevent the entrapment of air. A self-priming seawater pump shall be provided where pump lift requirements demand such a pump. Pumps that are not self-draining shall be provided with petcocks or other means of draining completely without the use of special tools (see 6.6.7). Engine manuals shall contain complete instructions covering draining of pumps. Seawater pumps shall be constructed of noncorrosive materials compatible with seawater. The use of cast iron in contact with seawater is not acceptable. Shaft seals for seawater pumps shall prevent any air from leaking into the pumps under a suction head of 381 millimeters of Hg (15 inches of Hg).

3.7.2.2 Engine fluid coolers. On-engine lubricating oil coolers, air charge coolers, and jacket water coolers shall be provided. When deemed necessary by the manufacturer, fuel coolers shall be provided.

3.7.2.2.1 Materials. Commercial cooler materials shall be as specified in tables I and II, except 70-30 copper-nickel alloy may be used where 90-10 is specified. The manufacturer shall have the option of substituting commercial material having equal or better physical and chemical properties than the materials specified in the applicable documents listed in tables I and II. It shall be the responsibility of the engine manufacturer to satisfy the cognizant Defense Contract Administration Service that this is the case.

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TABLE I. Seawater coolant.

Part	Material	Applicable documents
Tubes	Composition 90-10 copper-nickel alloy	ASTM B111/B111M
Shell hubs	Forged naval brass	ASTM B283/B283M, alloy 464
	Cast-bronze	ASTM B584, alloy 903 or 922
	Cast-brass	ASTM B584, alloy 857
Tube sheets 1/	Aluminum-bronze	ASTM B171/B171M, alloy 614 or 630
	Copper-nickel alloy	ASTM B171/B171M, alloy 706
	Wrought naval brass	ASTM B171/B171M, alloy 464
Water boxes	Forged naval brass	ASTM B283/B283M, alloy 464
	Cast-bronze	ASTM B584, alloy 903 or 922
	Cast-brass	ASTM B584, alloy 857
Shells	Brass tubing	ASTM B135/B135M
	Copper tubing	ASTM B75/B75M
Shell internals	Nonferrous	---
Bolts, studs, and nuts	Phosphor-bronze	ASTM B139/B139M
	Copper-silicon alloy	ASTM B98/B98M
	Aluminum-bronze, stress relieved, composition 5	ASTM B150/B150M, ASTM B169/B169M, alloy 614
Cap screws	Alloy steel	ASTM A320/A320M
Gaskets	Rubber, synthetic, cloth insertion	HH-P-151
	Non-asbestos, compressed	MIL-DTL-24696

NOTE:

1/ When design embodies a tube sheet integral with shell hub, materials listed for shell hubs may be used for tube sheets.

TABLE II. Freshwater coolant.

Part	Material	Applicable documents
Tubes	Admiralty metal	ASTM B111/B111M
Shell hubs	Forged naval brass	ASTM B283/B283M, alloy 464
	Cast-bronze	ASTM B584, alloy 903 or 922
	Cast-brass	ASTM B584, alloy 857
Tube sheets 1/	Aluminum-bronze	ASTM B171/B171M, alloy 614 or 630
	Wrought naval brass	ASTM B171/B171M, alloy 464
Water boxes	Forged naval brass	ASTM B283/B283M, alloy 464
	Cast-bronze	ASTM B584, alloy 903 or 922
	Cast-brass	ASTM B584, alloy 857

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TABLE II. Freshwater coolant – Continued.

Part	Material	Applicable documents
Shells	Brass tubing	ASTM B135/B135M
	Copper tubing	ASTM B75/B75M
Shell internals	Nonferrous	---
Bolts, studs, and nuts	Phosphor-bronze	ASTM B139/B139M
	Copper-silicon alloy	ASTM B98/B98M
	Aluminum-bronze, stress relieved	ASTM B150/B150M, ASTM B169/B169M, alloy 614
Cap screws	Alloy steel	ASTM A320/A320M
Gaskets	Rubber, synthetic, cloth insertion	HH-P-151
	Non-asbestos, compressed	MIL-DTL-24696

NOTE:

1/ When design embodies a tube sheet integral with shell hub, materials listed for shell hubs may be used for tube sheets.

3.7.2.2.2 Construction. Coolers shall be capable of withstanding a hydrostatic gage pressure of 0.0337 kg/m^2 (115 lb/in^2) on both the cooled and cooling side. Vent and drain connections shall be provided to permit complete venting and draining of both sides. Supports shall be provided for securing to a foundation. Coolers shall not be supported by plates or brackets in such a manner that the primary means of support is obtained from the bolts securing the shell end flange-tube sheet-water box flange joints.

3.7.2.2.2.1 Tubes. Coolers shall have straight tubes. Coiled or U-tubes are not permitted. Tubes shall be 6.350 millimeter (¼ inch) outside diameter (o.d.) by 0.559 millimeter (0.022 inch) thick wall (24BWG [Birmingham Wire Gage]) or 9.53 millimeter (⅜ inch) o.d. by 0.711 millimeter (0.028 inch) thick wall (22BWG). The minimum pitch for 6.350 millimeter (¼ inch) tubes shall be 7.938 millimeter (5⁄16 inch) and 11.509 millimeter (29⁄64 inch) for 9.525 millimeter (⅜ inch) tubes.

3.7.2.2.2.2 Tube joint. Tubes shall be either roller expanded into the tube sheets or attached to the tube sheets by silver-brazing. When roller expanded, the minimum effective depth of expansion shall be one tube in diameter. Provisions shall be made for the support of the tubes and prevention of tube vibration.

3.7.2.2.2.3 Cooling surface. The amount of cooling surface installed in all coolers shall be based on the service (specified in 3.7.2.2.3) applying a 10 percent fouling factor to the heat transfer coefficient for clean tubes.

3.7.2.2.3 Heat exchangers. Freshwater and seawater heat exchangers shall be water cooled and shall be in accordance with 3.7.2.2. The use of heat exchanger cores installed in expansion tanks or other enclosed areas in the cooling system is permitted provided the core construction and materials meet the requirements of 3.7.2.2.

3.7.2.2.4 Charge air cooler. When used, charge air coolers (also known as intercooler or aftercooler) shall be water cooled. The use of charge air cooler cores installed in intake manifolds or other enclosed areas of the air intake system is permitted provided the core construction and material requirements of the application meet the requirements of 3.7.2.2.

3.7.2.3 Water temperature control. Temperature regulator valves and thermostatic valves shall be either the fail-safe type that shall pass all the cooling water through the cooler in the event of failure of the thermostatic element or be provided with a manual override for the thermostatic element.

3.7.2.4 Inhibitors. Engines shall be capable of operating using coolant inhibitors and water treatment plans in accordance with S9086-GX-STM-030/220.

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3.7.2.5 Gasket and packing. Gaskets and packing used in the cooling system shall be of material resistant to deterioration by seawater or when any of the following are added to the freshwater system:

a. Antifreeze compounds in accordance with S9086-GX-STM-030/220.

b. Corrosion inhibitor compounds in accordance with S9086-GX-STM-030/220.

c. Preservative compounds in accordance with S9086-HB-STM-010/233.

3.7.3 Fuel system. Engines shall be capable of unrestricted, continuous operation at any power level consistent with the engine's load profile using naval distillate fuel F-76 (in accordance with MIL-DTL-16884), aviation turbine fuel grade JP-5 (in accordance with MIL-DTL-5624), #2 ultra-low sulfur diesel fuel (in accordance with ASTM D975), and DMA (in accordance with ISO 8217). The fuel system shall include the following:

a. Positive displacement type fuel pump

b. Relief valve connected to discharge into pump suction or supply tank

c. Priming pump

d. Filters

e. Necessary piping, valves, and fittings

3.7.3.1 Leak-off. If leak-off from the injectors and fuel pump exceeds 2 percent of the supply pump capacity, necessary equipment shall be provided to automatically return leak-off fuel to the service tank.

3.7.3.2 Fuel pump. The engine driven positive displacement fuel pump shall have a suction lift capability of 1.2 meters (48 inches).

3.7.3.3 Filtration. The fuel system shall be protected by a primary filter (strainer) ahead of the engine mounted fuel pump and a duplex secondary filter ahead of the fuel injection pump or unit injectors. If the fuel system is of such a design that the filter must be positioned on the suction side of the fuel pump, only the secondary filter is required. Filters shall be constructed of brass, bronze, or other metals that are not readily corroded by seawater contaminated fuels. Coated steel or untreated aluminum are not acceptable. Filters shall include vents to permit the expulsion of air. Manually operated drain valves shall be provided to completely drain filter bodies without the use of tools.

3.7.3.3.1 Primary filter. Primary filters (strainers) shall be the simplex metal edge (see 6.6.3) type with 0.0762 to 0.0889 millimeter (0.0030 to 0.0035 inch) spacing, enclosed manually operated cleaning device and sediment drain.

3.7.3.3.2 Duplex secondary filters. Duplex filters shall consist of two filter units connected to the common head or placed together equipped with a three-way plug valve, disposable filter elements, and mounting arrangement. The lever-operated valve shall direct flow through one side to permit element change and cleaning of the other side without interruption of flow. Each side shall be sized to accommodate twice the full flow of fuel supplied to the engine fuel manifold supply line or supply header at rated load of the engine and employ drain valves to permit complete draining. Filter elements shall be rated at 567.8 liters per hour (l/hr) (150 gallons per hour (gal/hr)).

3.7.3.4 Fuel oil priming pump. A hand-operated fuel oil pump shall be furnished with the capacity to prime the engine fuel oil system within 2 minutes at an ambient temperature of 21.1 °C (70 °F).

3.7.3.5 Minimum operating fuel viscosity. The manufacturer shall identify the minimum operating fuel viscosity so a determination can be made as to whether or not a cooler will be required within the fuel distribution system (see 6.2).

3.7.4 Lubricating system. Lubricating oil shall conform to S9086-H7-STM-010/262 for ambient condition above -6.7 °C (20 °F). The same grade of lube oil shall be used in the engine and attached accessories. Engines shall be provided with an oil level indicator, filling opening, and accessible sump drain connection or a sump pump to drain or clean the engine oil pan. Bayonet type oil level indicators shall be shielded to permit accurate readings while the engine is running. Engines shall be provided with suction and pressure connections for priming and flushing the engine lubrication system.

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3.7.4.1 Lubricating oil pumps. Pressure pumps, scavenging pumps, and piston cooling oil pumps shall be of the positive displacement type, driven by the engine. If required, the scavenging pump capacity shall be approximately 1¼ times the capacity of the pressure pump and shall have an oil flow capable of scavenging the system from idle to rated speed. A relief valve shall be provided for the outlet of each pump. If the pump suction incorporates a check valve, means shall be provided to prevent excessive build-up in the suction line in case the engine and pump are rotated backwards.

3.7.4.2 Lubricating oil coolers. Lubricating oil coolers shall be freshwater cooled and shall be in accordance with 3.7.2.2. The use of cooler cores installed in an enclosed area integral to the engine is permitted provided the core construction and materials meet the requirements of 3.7.2.2.

3.7.4.3 Lubricating oil filters. Engines shall use full-flow filter elements conforming to type IV. Full-flow filter relief valves shall be external to the filter case. Valves shall bypass oil when the pressure gage drop reaches the predetermined setpoint. The valve closing gage pressure shall be not less than 0.0059 kg/m^2 (20 lb/in^2). The filter inlet shall be internally baffled or so constructed to prevent the oil flow from impinging directly on the filter elements. The filter shall include a hand-operated drain valve, which shall completely drain the filter for servicing. Automatic or backflushing filters shall have a separate filter element to filter the oil returning to the sump.

3.7.4.4 Lubricating oil strainers. Strainers shall be either the simplex metal edge type or the duplex metal edge or basket type. The strainer shall be sized to carry the full flow of 150 standard saybolt universal (SSU) viscosity oil with a pressure gage drop of not more than 0.0015 kg/m^2 (5 lb/in^2) with clean elements. The strainer bypass valve shall be set to open within the differential gage pressure range of 0.0059 to 0.0067 kg/m^2 (20 to 23 lb/in^2) across the element and shall be capable of passing the full oil flow with a total gage pressure drop through the bypass of less than 0.0076 kg/m^2 (26 lb/in^2). Strainers shall include hand-operated drain valves, which shall permit complete drainage of the body(s) for servicing.

3.7.4.4.1 Simplex strainer. Simplex strainers shall be the metal edge type with 0.127 millimeter (0.005 inch) spacing and an enclosed manually operated cleaning device. Wire cloth, screen, sintered metal, or perforated plate media shall not be used for simplex strainers.

3.7.4.4.2 Duplex strainer. Duplex strainers shall consist of two filtering units connected by a changeover valve. The arrangement shall employ a common head with an integral valve or separate units piped together with an external valve. The changeover valve shall direct the oil flow through one side to permit cleaning of the other side without interruption of flow. Each side shall be sized to accommodate the entire lubrication oil flow to the engine. Filter elements shall be either the metal edge type, with or without a manual cleaning device, or the wire screen basket type. Wire screen media shall be 100 mesh or smaller and shall be protected on both sides to prevent deformation at differential gage pressures up to 0.0205 kg/m^2 (70 lb/in^2).

3.7.4.5 Lubricating oil priming pump. A lubricating oil pump shall be furnished capable of priming the engine lubricating oil system within 2 minutes at an ambient temperature of 21.1 °C (70 °F).

3.7.4.6 Lubricating oil preheat. Engines shall be equipped with a lubricating oil heating system to maintain the lubricating oil at a temperature of 43.3±5.5 °C (110±10 °F) when the engine is secured. The system shall employ forced circulation to distribute the oil through the external lubricating oil piping and the engine oil sump. The heating system shall be equipped with both manual and automatic shutoff devices. The automatic shutoff shall secure the system in the event of a loss of engine lube oil or an engine start. This system may be combined with the lubricating oil priming pump in 3.7.4.5.

3.7.5 Air intake systems. The combustion air shall be taken from the engine compartment or from ducts provided by the installing activity, as specified (see 6.2).

3.7.5.1 Filtration. When specified that the engine will take its combustion air from the engine compartment (see 6.2), the engine shall be provided with an effective air intake silencer and with a cleanable filter or screen necessary to protect the equipment from ingesting dirt. Oil bath filters or media that would support combustion or produce toxic fumes during a fire in the engine space shall not be used. Use of media other than a metallic type or flameproof paper requires Technical Authority approval. Filters shall incorporate a device that shall indicate when the element requires changing or cleaning. The intake system shall be directed or extended as necessary to avoid unnecessary heat transfer to combustion air from hot engine parts such as exhaust manifolds and turbocharger turbine casings.

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3.7.5.2 Ducts. When specified that the engine will take combustion air from ducts provided by the installing activity, the following equipment shall be furnished with the engine and be installed as described:

a. An air filter shall be installed close to the engine.

b. The filter elements shall not deform or collapse with the engine operating at full-rated speed and load at the maximum intake depression condition.

c. The area of the filter element shall be such that the air velocity at the face of the filter does not exceed 2,000 feet per minute (ft/min), or the pressure drop across the clean filter be more than 101.6 millimeters of water (4 inches of water) at rated output.

d. A manometer or signal device shall be connected across the element to indicate when cleaning is required.

e. The filter housing shall permit the removal of the element for cleaning.

f. The filter element shall be installed at an angle of 90 degrees or less to the direction of air flow in order that impinging dirt will drop to the bottom of the duct on the upstream side of the filter element.

3.7.5.3 Noise level requirements. Enclosure noise abatement systems shall be fitted, if necessary, to limit unmanned machinery space noise levels and manned machinery noise levels to those specified (see 6.2). Measurements and instrumentation shall be in accordance with MIL-STD-740-2 and MIL-STD-1474.

3.7.5.4 Blowers and air receivers. Means shall be provided for draining pockets in the blower housing and air receiver where oil or water may accumulate.

3.7.6 Exhaust system. Exhaust headers shall be freshwater cooled, insulated, or shielded. Asbestos insulating products and materials shall not be used. Provisions shall be made for draining the gas spaces or the header from either end. When required to be furnished with engines (see 6.2), exhaust mufflers shall be in accordance with 3.7.6.1.

3.7.6.1 Exhaust mufflers. Enclosure noise abatement systems shall be fitted, if necessary, to limit unmanned machinery space noise levels and manned machinery noise levels to those specified (see 3.7.5.3).

a. Exhaust mufflers shall be made of corrosion resistant materials. Weathering steel shall not be used.

b. The muffler shall be the dry type with a spark arrestor. All-metal parts shall be constructed of carbon steel. Drains for removal of water and oil and a soot collector for removal of carbon particles shall be included.

3.7.6.2 Exhaust emission. Diesel engines shall comply with the emissions requirements applicable to marine engines intended for United States (U.S.) Flag commercial ships.

a. CFR; Title 40, Protection of Environment. oxides of nitrogen (NO_X), carbon monoxide (CO), hydrocarbon (HC), particulate matter (PM) and smoke emissions shall conform either to the applicable regulations in Part 89 – Control of Emissions from New and In-use Non-road Compression-Ignition Engines or Part 94 – Control of Air Pollution from Marine Compression-Ignition Engines.

b. MARPOL Annex VI requirements. Unless otherwise specified (see 6.2), NO_X emissions shall conform to MARPOL Annex VI, Regulation 13.

3.7.7 Starting systems. The diesel engine installation shall be provided with a system for starting each diesel engine. The starting system shall be able to start the diesel engine from normal ambient condition and cold start condition and, when specified (see 6.2), from submerged condition and emergency condition. Options for diesel engine start systems are:

a. Air starting

b. Hydraulic starting

c. Electric starting

3.7.7.1 Air starting. The air starting system shall consist of the starting motor or distributor, inlet relay valve, manually operated control valve and the necessary lines, connections, and fittings. The system shall include a device to automatically shut off the starting air when the engine starts.

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3.7.7.2 Hydraulic starting. The hydraulic starting system shall consist of the starting motor, accumulator(s), engine-driven charging pump, hand-operated charging pump, reservoir, control valve, dumping valve, pressure gage, filter, and the necessary lines, connections, and fittings. The system shall be designed to be self-contained and non-dependent on any other ship system. Interconnecting lines and fittings will be furnished by the installing activity. The following pressure ratings shall apply:

a. Working pressure – 2.0684±0.069E+7 Pascals (Pa) (3,000±100 pounds per square inch (psi))

b. Accumulator proof pressure – 4.1369E+7 Pa (6,000 psi)

c. Accumulator burst pressure – 8.2737E+7 Pa (12,000 psi)

d. Accumulator fuse blow-out pressure – 3.2405±0.2069E+7 Pa (4,700+300 psi)

e. Hose and fitting proof pressure 3.1026E+7 to 8.2737E+7 Pa (4,500 to 12,000 psi)

f. The engine-driven charging pump shall be capable of charging the system from zero pressure to the working pressure within 20 minutes with the engine running at rated speed. After the system is fully charged, the engine-driven charging pump and pressure regulator shall maintain the system at the specified working pressure. The hand-operated charging pump(s) shall be capable of fully charging the system from zero pressure to the working pressure within 20 minutes at temperatures between 10 and 32 °C (50 and 90 °F).

g. Accumulators shall be of a type that will quickly relieve the pressure when the shell becomes initially distorted as a result of impending rupture. Accumulators shall be mountable in either a horizontal or vertical position and shall contain a blow-out fuse. Accumulators shall be capable of being connected in parallel and shall have sufficient oil volume to meet the cold start test requirements (see 4.2.7.2.2). All accumulator shells shall be liquid penetrant and radiographically inspected in accordance with MIL-STD-271, prior to assembly. Acceptance standards shall be in accordance with MIL-STD-2035. The accumulator shall be made of a design, material, and construction that has been approved by the U.S. Coast Guard. Accumulators identical to a design previously approved do not need to be individually certified.

3.7.7.3 Electric starting. The electrical starting system shall be 24-volt, ungrounded, two-wire consisting of starting motor, contactor, battery charging alternator, voltage regulator, ammeter, and starter button. The electrical equipment shall be fungi- and corrosion-resistant. Starting batteries are not required unless specified (see 6.2).

3.7.7.3.1 Starting motor. The shifting mechanism of the starting motor shall be the totally enclosed, solenoid-actuated overrunning clutch type that will provide positive pinion and ring gear engagement before the power is applied. It shall not disengage due to intermittent engine starting and shall prevent the engine from driving or overspeeding the motor.

3.7.7.3.2 Battery charging unit. The battery charging unit shall consist of a three-phase alternating current (AC) generator, rectifier, and control unit.

3.7.7.3.3 Ammeter. The battery charging generator ammeter shall be in accordance with MIL-PRF-10304. The meter shall register discharge to the left of zero and charge to the right of zero. The meter shall be self-contained with a 63.5 millimeter (2½inch) diameter. It shall have a flush-mounted luminous dial, a wide flange, and a metal case with a transparent heat-resistant window. The meter shall be hermetically sealed. The rear of the case shall be provided with conventional terminals. Zero correction is not required.

3.7.8 Speed governing system. Diesel engines driving propulsion, ship service or emergency generators, and auxiliary equipment shall be equipped with a speed governing system capable of automatically maintaining the required engine speed for that particular application. The engine shall also be equipped with an independently-driven overspeed protective device.

3.7.8.1 Parallel operation. Engines that are required to operate in parallel on a single propulsion shaft system shall be provided with a load limiter or fuel stop, which will automatically control engine torque to a safe limit. Regulating governors shall be provided with a steady-state speed regulation mechanism, adjustable from 0 to 5 percent of rated speed. Governors for generator drive engines required to parallel shall be equipped with a reversible synchronizing motor when control is from a remote operator’s station.

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3.7.8.2 Electronic speed governing systems. Digital electronic governing systems for variable speed engine operation shall be provided with dual redundant engine control modules. Governor response shall not be adversely impacted by the natural frequency of the engine coupling. The governor shall be adjustable from idle speed to a minimum of 5 percent above the rated speed. The back-up control module shall automatically take control of the engine upon failure of the primary control module. Indicators shall be provided to alert the operator upon loss of redundancy. When electronic speed governors are fitted to main propulsion diesel engines or generator diesel engines and form part of the remote propulsion control system, they shall comply with the following:

a. If lack of power to the governor control and associated hardware may cause major and sudden changes in the preset speed or direction of thrust of the propeller, an automatically available self-contained backup power supply shall be provided so as not to interrupt the power supply to these systems. An alarm for the failure of the main power supply shall be provided at the main and remote (if provided) propulsion control stations.

b. Local control of the engines shall be possible. For this purpose, means shall be provided at the local control position to disconnect the remote-control signal. If this will disconnect the speed governing functions required by this specification, an additional separate speed governing system shall be provided for such local mode of control.

c. When operating at all loads up to the rated load, the speed for any constant load shall be so controlled that the periodic or aperiodic oscillations of speed shall be no greater than ±1 percent of rated speed.

d. The response of the electronic governing system to sudden increase or decrease in load shall meet the requirements of ISO 3046-4 for a type 3, class A1 application. In addition, the engine revolutions per minute (RPM) shall return to and remain within ±1 percent of the final steady-state speed in not more than 15 seconds following a change in load.

e. For multiple propulsion diesel engines coupled to a single propulsion shaft, the electronic speed governing systems shall provide for a means to balance the load between all engines on that shaft to ensure engine overload thresholds are not reached.

3.7.8.3 Generators in parallel. For diesel engines driving AC generators that operate in parallel, the governor's characteristics shall be such that in the range between 20 and 100 percent of the combined rated load of all generators, the load on any individual generator will not differ from its proportionate share of the total combined load by more than 15 percent of the rated power of the largest generator or 25 percent of the individual generator, whichever is less. Diesel engine driven generator sets shall be fitted with a governing system that will allow for both Isochronous and Droop modes of operation.

a. Isochronous mode: When parallel operation is required, real load differential compensation shall be performed in order to maintain isochronous frequency regulation and real load balance.

b. Droop mode: Steady-state speed shall be adjusted in the negative direction with an increasing kW load by the droop control circuit. Droop mode shall provide stable operation with dissimilar governors, infinite bus, or another generator set with the same governor operated in Isochronous mode. Frequency droop at 100 percent load shall be adjustable from 0 to 5 percent of the no load frequency in increments no greater than 0.2 percent.

3.7.8.4 Overspeed protection. As protection against dangerous engine overspeed due to failure of the regulating governor, its linkage, or other engine malfunction, each engine shall be equipped with a separate and independent speed limiting device that will control the engine when it attempts to exceed the overspeed setpoint of its full-rated speed for the intended application. The speed limiting device shall take control at overspeed and stop the engine completely by reducing the fuel injection output to zero or blocking the engine's combustion air supply, or both. This device shall require manual resetting. The speed limiting device shall not attempt to control the engine by any system that cuts off the fuel supply to the engine fuel pump. Governor and overspeed devices shall be tested in accordance with 4.2.7.3 and 4.2.7.13.

3.7.8.5 Low lubrication oil pressure protection. The engine shall be provided with a device that will automatically shut down the engine when the lube oil pressure drops to a point that will cause failure of the oil-wetted wearing parts. The device shall actuate at a pressure selected by the engine manufacturer to ensure protection of the engine. The shutdown device shall be equipped with a manual override.

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3.7.9 Instrumentation. Instruments shall be furnished with each engine for the following measurements:

a. Pressure indicators:
   (1) Freshwater pump discharge
   (2) Seawater pump discharge
   (3) Lubricating oil pressure pump discharge
   (4) Lubricating oil pressure at the engine inlet
   (5) Piston cooling oil (if separate pumps are used)
   (6) Fuel oil pump discharge
   (7) Fuel oil pressure at injection pump inlet or fuel header
   (8) Scavenging or supercharging air pressure
   (9) Starting air pressure at the engine
   (10) Lubricating oil pressure at the turbocharger or blower
   (11) Crankcase pressure/vacuum
   (12) Scavenging/supercharger air after any inline heat exchangers

b. Temperature indicators:
   (1) Lubricating oil from engine
   (2) Freshwater from engine
   (3) Water from the intake air cooler (if used) (see note below)
   (4) Seawater to and from heat exchanger (see note below)
   (5) Freshwater to lubricating oil cooler (see note below)
   (6) Freshwater to the engine (see note below)
   (7) Lubricating oil to engine (see note below)
   (8) Exhaust before and after turbocharger
   (9) Air intake before and after heat exchangers
   (10) Electrical pyrometer and selector switch, with one thermocouple at each cylinder exhaust outlet and exhaust side of turbocharger(s)

Note: Temperature indicators will be furnished for installation in ships piping by the installing activity.

c. Alarms:
   (1) Low-lubricating oil pressure
   (2) High-lubricating oil temperature from the engine
   (3) High-temperature cooling water
   (4) High-crankcase pressure
   (5) High-intake manifold air temperature (turbocharged engines only)

d. Tachometer: Engine speed

e. Time indicator or hour meter: Engine operating time

3.7.9.1 Panel. Instruments shall be furnished with the engine for local engine monitoring containing all necessary instrumentation mounted on a local instrumentation panel on or adjacent to the engine.

3.7.9.2 Alarms. A low-lubricating oil pressure alarm switch, connected to the remote end of the lubricating oil system, and a cooling water high-temperature alarm switch with an actuation bulb installed in the cooling water outlet, shall be provided. The low-lube oil pressure alarm shall be set higher than the setting established for the automatic low-lube oil pressure engine shutdown (see 3.7.8.5).

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3.7.10 Engine-mounted control system. Unless otherwise specified (see 6.2), the engine shall be provided with attached controls for starting, stopping, and speed regulation. Controls necessary for operation of the engine shall be conveniently grouped at one end of the engine. The governor fuel actuating shaft or fuel injection pump control linkage shall be provided with a pointer and a graduated scale to indicate pump rack position.

3.7.11 Emergency shutdown device. Engines shall be provided with an emergency shutdown device operable both on the engine and by a pull cable, or other means from a remote location, which shall secure fuel flow to the injectors, shut off the combustion air, and stop the engine without any damage within 60 seconds. The device shall be of a type that requires manual resetting before the engine can be restarted. When the engine takes combustion air from the engine compartment the shutdown device shall, at a minimum, shut off the air to prevent ingestion of the fire suppressant medium. This device may be combined with the speed limiting device (see 3.7.8.4). The remote shutdown device will be furnished by the installing activity.

3.7.12 Interference shielding. Electrical equipment and interconnections shall be interference shielded in accordance with S9407-AB-HBK-010.

3.7.13 Resiliently-mounted equipment. Where resilient mounts are used, they shall be within the limits of the manufacturer's instructions and specifications for the resilient elements' stiffness (both static and dynamic), elasticity, noise suppression, shock and vibration capabilities, overturning moments due to output torque and durability under shipboard ambient conditions. Replacement and maintenance frequency and procedures shall be provided. Resilient mounts should maintain engine alignment requirements in all operating conditions including in the event of mount failure.

3.7.14 Torsional vibration. A torsional vibration analysis shall be conducted in accordance with MIL-STD-167-2, with the exception of "Torque across gears", "Simulated installation tests", and "Auxiliary branches".

3.7.15 Engine turning or jacking device. A device for barring the engine over, arranged for power and manual operation, shall be provided. An interlock shall be provided to prevent the engine from starting when the barring device is engaged. A flywheel incorporating a jacking gear shall be provided with each engine. Alternatively, a gear train arrangement to the engine, independent of the flywheel, may be furnished. Each jacking gear shall have a manually operated engaging and disengaging pinion driven by a reversible electric or pneumatic motor jacking device which will permit accurate adjustment of crank angles for the purpose of engine inspection and maintenance. The jacking gear shall be equipped with a control extension cord of sufficient length to be capable of controlling operations from all engine levels and at each crankcase door and each cylinder head. The flywheel or engine crankshaft flange shall be provided with a degree scale and marking to permit checking crank positions.

3.7.16 Cylinder test and indicator cocks. Cylinder test and indicator cocks shall be provided if available on the engine design.

3.8 Low load testing. When specified (see 6.2), the diesel engine shall be capable of withstanding low load testing in accordance with 4.2.7.19.

3.9 Painting. The engine shall be painted. Internal oil-wetted surfaces shall not be painted.

3.10 Designating and marking. Engines, crankcases, crankcase main bearings, components, and parts shall be marked for identification in accordance with MIL-STD-130. Repair parts shall be marked with the manufacturer's part number and serial number by casting, forging, stamping, or etching. When it is impractical to mark repair parts by any of these methods, the parts shall be identified by tags or a printed sheet on or in the package.

3.10.1 Engine. Engines shall be provided with an identification plate secured to the engine in a visible and convenient location. Identification plates shall be of metal and shall show the following:

a. Engine type
b. Serial number
c. National stock number
d. Contract number
e. Model number
f. Rated speed (RPM)

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g. Brake horsepower (BHP)

h. Rotation direction as viewed from the drive end

i. Bore

j. Stroke

k. Dry weight

3.10.2 Navy identification plate rating. The HP rating indicated on the engine identification plate shall be the engine full-power output. In no case may the engine identification plate HP be greater than the rated (full) load used in first article testing (see 4.2.6).

3.10.3 Accessory components. Complete accessory components such as pumps, coolers, filters, governors, electrical, and starting equipment shall be provided with identification plates showing the manufacturer's name, model or part number, and capacities or ratings where applicable.

3.10.4 Rotation. A plate with an arrow showing the direction of rotation (see 6.6.5) shall be attached to the major output end of the engine.

3.10.5 Spare parts. Spare parts shall be marked with the manufacturer's part number by casting, stamping, or etching. Where it is not practical to mark parts by any of these methods, the parts shall be identified by tags or printed description on the package.

3.10.6 Valves. Valves are to be provided with the following information: name and address of manufacturer, designation and size, month and year of manufacture, and approved installation orientation.

3.11 Drawing. When specified (see 6.2), installation and accessory drawings shall be provided.

3.12 Manuals. When specified (see 6.2), manuals shall be provided.

3.13 Workmanship. Castings shall be free from sand, shrinks, cold shuts, cracks, harmful porosity, and other defects that would cause premature failure. Runners, risers, fins, and other cast-on pieces shall be removed. Sharp edges on projections shall be removed from stamping or forgings in the finished part. Machined surfaces shall have sharp edges broken or chamfered. Weld shall be free of weld spatter or slag. Forgings shall be free from seams, cracks, scale, fins, porosity, hard spots or excessive inclusions, segregations, or other defects.

4. VERIFICATION

4.1 Classification of test and inspections. The test and inspection requirements specified in table III are classified as follows:

a. First article test and inspection (see 4.2).

b. Conformance test and inspection for both package and component (see 4.3).

TABLE III. First article and conformance tests and inspections.

Procedure	Requirement	Test	First article	Conformance	
				Package	Component
Standard 8-hour performance cycle	3.1	table VI	X	X 1/	X 1/
Shock testing	3.4.2	4.2.7.18	X	----	----
Engine alignment	3.4.3	4.2.7.9	X	X	----
Compression and firing pressure	3.4.4	4.2.7.10	X	X 2/	X 2/
Piping integrity	3.7.1	4.2.7.4	X	----	----
Waste heat	3.7.2	4.2.7.5	X	----	----
Jacket water analysis	3.7.2.5	4.2.6.6.4	X	----	----

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TABLE III. First article and conformance tests and inspections – Continued.

Procedure	Requirement	Test	First article	Conformance	
				Package	Component
Fuel consumption	3.1.3	4.2.7.11	X	----	----
Fuel performance comparison	3.7.3	4.2.7.12	X	----	----
Fuel oil analysis	3.7.3	4.2.6.6.1	X	X	X
Lubricating oil consumption	3.7.4	4.2.6.6.2	X	----	----
Lubricating oil analysis	3.7.4	4.2.6.6.3	X	X	X
Airborne noise	3.7.5.3	4.2.7.7	X	----	----
Structureborne noise	3.7.5.3	4.2.7.8	X	----	----
Exhaust smoke emissions	3.7.6.2	4.2.7.14	X	X	X
Normal (ambient) starting	3.7.7	4.2.7.2.1	X	X	X
Cold starting	3.7.7	4.2.7.2.2	X	----	----
Emergency starting	3.7.7	4.2.7.2.3	X 3/	X 3/	X 3/
Submerged starting	3.7.7	4.2.7.2.4	X 4/	----	----
Governor response	3.7.8.2	4.2.7.3	X	X	X
Overspeed governor	3.7.8.4	4.2.7.13	X	X	X
Alarms and shutdown	3.7.9	4.2.7.15	X	X	X
Emergency shutdown	3.7.11	4.2.7.1	X	X	X
Torsional vibration	3.7.14	4.2.7.6	X	----	----
Low load testing	3.8	4.2.7.19	X	----	----
Visual and dimensional analysis	3.13	4.2.3.1 and 4.2.6.10.1	X	----	----
Reassembly	3.13	4.2.6.10.2	X	----	----

NOTES:

1/ Only one 8-hour cycle is required for conformance testing.

2/ Where engine is equipped with measuring capability.

3/ Where applicable (for emergency diesel generator [EDG] applications).

4/ Where applicable (for submerged applications).

4.2 First article test and inspection. First article test and inspection shall be performed on one production representative complete engine or engine package, when a first article sample is required (see 3.1), in accordance with table III. First article diesel engines shall be subject to comprehensive testing to include operating endurance testing, to include first article pre- and post-test examinations and tests and first article specific tests (see table III), some of which are also applicable to conformance testing (see 4.3). This inspection shall include the tests of 4.2.1 through 4.2.7.

4.2.1 Type test. A type test carried out for a type of engine at a place of manufacture will be accepted for all engines of the same type that are built by both licensors and licensees of the technology. A type test carried out on one engine having a given number of cylinders will qualify all engines of the same type having a different number of cylinders.

4.2.2 Government witness. When specified (see 6.2), the Government shall be allowed to witness testing.

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4.2.3 First article test conditions.

4.2.3.1 Visual and dimensional examination. The engine package shall be subjected to a visual and dimensional examination to ensure that the material, design, workmanship, surface finish, deposit level, and construction areas specified herein not involving disassembly or tests. This visual and measurement examination along with as-built drawings shall establish the configuration baseline for the component evaluation after endurance tests. Engine parts shall not be replaced or exchanged without prior approval from the Technical Authority. The engine and its attached and unattached assemblies shall be disassembled, each piece shall be examined, and measurements shall be made before and after the first article test. Special tools furnished with the engine shall be used during this examination. This examination shall be conducted on all components replaced during the first article tests. All parts to be replaced as part of maintenance scheduled during testing shall also be identified prior to the start of testing. The visual examination checklist shall include surface defects on un-machined surfaces, scuff marks, unusual deposits, discoloration, abnormal wear, fretting, beginning of pitting, and cavitation.

4.2.3.2 Measurement standards. The wear limits or comparison photographs for all wearing engine components, variability among the parts of the same part number, dimensional acceptance limits, variability limits, limits for all running clearances and adjustments, design tolerances and clearances, engine operating parameters, maximum loads for speeds between idle speed and rated speed, lubricating oil and jacket water spectrographic trending analysis limits, as applicable, shall be submitted for approval to the Technical Authority prior to the start of testing when specified (see 6.2).

4.2.3.3 Evaluation criteria. The failure of any component to meet the manufacturer's wear limits or evidence of cracks, cavitation, pitting, scuffing, galling, fretting, or overheating shall be cause for the rejection of the engine. A failure to meet evaluation criteria shall be cause for rejection of the engine. The engine's failure to meet requirements shall result in a restart of the first article tests.

4.2.3.4 First article test operating conditions. Unless otherwise specified (see 6.2), the engine shall be subjected to the operating conditions listed in 3.3.1.

4.2.3.5 Restriction settings. Inlet air and exhaust system restrictions shall be measured at locations in accordance with ISO 3046-1. The first article test intake air depression and exhaust back pressure restriction values specified in 3.3.1 shall be set by throttling for all first article tests. The specified restrictions values shall be set at corrected rated load and speed. The settings shall remain unchanged during all specified first article tests.

4.2.3.6 Test fluids. The fuel used during the tests shall be naval distillate fuel (F-76) in accordance with MIL-DTL-16884 with exception of the tests noted to be run with JP-5 in accordance with MIL-DTL-5624. When specified (see 6.2), some or all of the other tests and examinations (see 4.2.7) shall be tested with F-76 or JP-5. Lube oil used during the tests shall be in accordance with the engine manufacturer's recommendations with approval by the Technical Authority. The jacket water shall be treated with corrosion inhibitor as recommended by the engine manufacturer with approval by the Technical Authority. Requests to use commercially-available alternative fuels during testing shall be subject to approval by the Technical Authority (see 6.2). Alternative fuels must demonstrate physical and chemical properties similar to that specified by MIL-DTL-16884 or MIL-DTL-5624 as applicable.

4.2.3.7 Commencing or recommencing. Upon activation and stabilization of the lubricating-oil preheat system, the jacket water keep-warm system, and the lubricating-oil priming system, the engine shall start and attain the first step or operating condition of each 8-hour endurance cycle within the time specified in table IV. Request to use longer ramp time than that specified in table IV shall be subject to approval by the Technical Authority (see 6.2).

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TABLE IV. Commencing or recommencing testing.

Rated power	Time to attain first step or operating condition	Time to attain second step or commencing or recommencing operating condition
Propulsion:		
≤ 1000 kW (1341 HP)	10 seconds	10 seconds
1000 kW (1341 HP) ≤ 3000 kW (4023 HP)	20 seconds	20 seconds
3000 kW (4023 HP) ≤ 5000 kW (6705 HP)	30 seconds	20 seconds
> 5000 kW (6705 HP)	40 seconds	20 seconds
Ship's Service Generator or Emergency Generator:		
≤ 1000 kW (1341 HP)	10 seconds	10 seconds
1000 kW (1341 HP) ≤ 3000 kW (4023 HP)	20 seconds	20 seconds
> 3000 kW (4023 HP)	30 seconds	20 seconds

4.2.3.8 Engine monitoring. Engine operation shall be monitored throughout all of the tests. Specified parameters shall be measured every hour at every operating condition (except 10-minute idle conditions) after the engine has been operating at a particular condition for at least 10 minutes and the temperature and pressures have stabilized. Monitored parameters shall be measured every hour and prior to reducing power for the idle speed steps and the shutdowns. Test events shall be monitored. Charge air cooler performance shall be monitored and verified within limits. The high temperature and low temperature pre-heating system performance shall be within limits.

4.2.3.9 Engine modification. From the start of the test, no parts shall be modified, replaced, or adjusted, except for scheduled maintenance or to correct permissible shutdowns (see 4.2.3.10.2).

4.2.3.10 Engine shutdowns. Engine shutdowns during the endurance test shall be defined as either permissible shutdowns or forced shutdowns.

4.2.3.10.1 Shutdown conditions. The engine shall be shut down if one or more of the following shutdown conditions occur:

a. Abnormal operation or failure of an engine component

b. Engine misfire during any test cycle operating condition

c. Engine operation outside of operating parameters as defined in the test procedure (see 6.2)

d. Engine torque or speed exceeding acceptable range of variability as defined by the engine manufacturer and agreed upon by the Technical Authority

e. Lubricating oil exceeding any physical test trending limit as defined by the engine manufacturer and agreed upon by the Technical Authority

f. Test personnel or engine endangered

g. Faulty test cell service systems or other systems extraneous to the engine

4.2.3.10.2 Permissible shutdowns. Permissible shutdowns include the minimum 10-minute shutdown period at the end of each cycle, shutdowns due to facility problems, and shutdowns at the end of the test day. Scheduled maintenance shall be performed during this shutdown period, which may be lengthened as desired by the manufacturer to carry out the required maintenance established in the approved first article test procedure (see 6.2). A shutdown shall be classified as permissible only if it cannot be described by any of the forced shutdown criteria listed in 4.2.3.10.3.

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4.2.3.10.3 Forced shutdowns. An engine shutdown shall be classified as a forced shutdown if it can be described by one or more of the following:

a. A single failure of an engine component that cannot be corrected or repaired within a 2-hour period (if the engine shutdown occurred during off-hours, initiation of the repair may be delayed until the start of the next workday). The 2-hour period does not include obtaining replacement parts, tools, or personnel to perform the repair. The 2-hour period is to include the removal of all interferences and draining of fluids from pipes that need to be disassembled once the work begins.

b. Two failures of the same engine components with the same part number. (The failure of identical components in different locations on the engine is classified as a forced shutdown.)

c. Engine repair requires access to the internal crankcase components or gear train; or requires the removal of manifolds, blowers, turbochargers, or cylinder heads.

d. Failure to maintain engine operation within the variability limits of the operating parameters, target torque, or target speed; or failure to maintain lubricating oil from exceeding the physical test trending limits.

4.2.3.11 Compression pressures. Where engine is equipped, compression pressures for each cylinder at idling speed, and firing pressure at rated-load and speed shall be measured and recorded before and after endurance tests and submitted with the first article report (see 4.2.8 and 6.3.6).

4.2.3.12 First article test plan. A first article test plan shall be prepared as specified (see 6.2).

4.2.3.12.1 First article inspections and tests. The inspections and tests to be conducted before, during, and after the first article test, including the endurance cycles, are listed in table V.

TABLE V. First article inspections and tests.

Paragraph	Inspection or Test
Before first article test:	
4.2.7.1	Emergency Shutdown
4.2.7.2	Engine Starting
4.2.7.3	Governor Response
4.2.7.4	Piping Integrity
4.2.7.5	Waste Heat
4.2.7.6	Torsional Vibration
4.2.7.7	Airborne Noise
4.2.7.8	Structureborne Noise
4.2.7.9	Alignment
4.2.7.10	Compression and Firing Pressure
4.2.7.11	Fuel Consumption
4.2.7.12	Fuel Performance Comparison
4.2.7.13	Overspeed Governor
4.2.7.15	Alarms and Shutdowns
During (concurrent) first article test:	
4.2.6	Standard 8-hour Endurance Cycle
4.2.6.6.1	Fuel Oil Analysis
4.2.6.6.2	Lubricating-Oil Consumption

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TABLE V. First article inspections and tests – Continued.

Paragraph	Inspection or Test
During (concurrent) first article test (Continued):	
4.2.6.6.3	Lubricating-Oil Analysis
4.2.6.6.4	Jacket-Water Analysis
4.2.7.14	Exhaust Smoke Emission
After first article test:	
4.2.6.10.1	Visual and Dimensional
4.2.6.10.2	Reassembly
4.2.7.1	Emergency Shutdown
4.2.7.9	Alignment
4.2.7.10	Compression and Firing Pressure
4.2.7.11	Fuel Consumption
4.2.7.13	Overspeed Governor

4.2.4 Generator set testing. The first article inspection and tests shall be in accordance with the requirements for main propulsion diesel engines, as specified in 4.2, with the exception of the following modifications:

a. Engine-to-generator alignment limits shall be determined and measured

b. Exhaust emissions shall be measured using test cycle “D2” as outlined in ISO 8178-4 and in accordance with 4.3.1

c. Monitoring during testing shall include the displays and alarms in the Local Control Panel

d. The standard endurance test shall consist of the steps described below and performed in the order given therein

e. Examinations and tests for application as a ship service generator set listed in 4.2.4.1 through 4.2.4.5

4.2.4.1 Examinations. In addition to the first article tests specified in 4.2, the generator set shall be examined for access to accessories, such as oil sight flow gauges, lube oil fill and drain, air and oil filter, instruments, and so forth. The fit of parts shall be observed with particular reference to the interchangeability of such parts as are likely to require replacement during the normal service life of the generator set.

4.2.4.2 Packaging test. When a previous Navy-qualified propulsion diesel engine that does not require any testing is proposed for a main propulsion generator set, ship service generator set, or auxiliary service set application, the first engine shall undergo the packaging test requirements outlined in the list below.

4.2.4.3 Standard 8-hour endurance cycles. Twelve standard 8-hour endurance cycles, composed as specified in table VI, shall be observed in this test.

a. Measurements of observed load shall be corrected using the power correction method (see 6.2).

b. Monitoring during testing shall include the indications and alarms on the Generator Control Unit (Control Panel).

c. Measurements of monitored parameters shall be made at intervals specified in the engine monitoring requirements of the first article inspection.

d. Lubricating oil samples shall be collected before and after the cycle and analyzed as specified in the first article inspection and tests.

e. Exhaust smoke and emissions shall be measured using the methods specified for the first article testing at each operating point of the standard 8-hour endurance cycle. Exhaust emissions shall comply with requirements set forth in first article emission testing.

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4.2.4.4 Generator testing. Testing of the generator set shall include all parts being supplied by the manufacturer for installation with the generator set such as the governing system and shall ensure compliance with MIL-DTL-3124. Standard commercial test cells, detached accessories, and instrumentation may be used for these tests.

4.2.4.5 Test criteria. A failure during the test cycle shall be cause for rejection. A repaired engine shall be retested for one cycle of the standard 8-hour endurance test. The first article test requirements listed below shall also apply to the packaging test.

a. Commencing (see 4.2.3.7)
b. Engine monitoring (see 4.2.3.8)
c. Engine modification (see 4.2.3.9)
d. Engine shutdowns (see 4.2.3.10)
e. Permissible shutdowns (see 4.2.3.10.2)
f. Forced shutdowns (see 4.2.3.10.3)
g. Test profile completion (see 4.2.6.9)
h. Inspections after completion of test profile (see 4.2.6.10)

4.2.5 First article inspection and testing requirements for engines rated 375 kW (500 HP) or less. Diesel engines with a rated power of 375 kW (500 HP) or less and currently produced for commercial applications shall be subject to first article inspection and testing as modified in 4.2.5.1 through 4.2.5.6.

4.2.5.1 Load points.

a. 80 hours at rated power
b. 8 hours at 110 percent overload and, alternately, 100 percent RPM and 103 percent RPM
c. 10 hours at partial loads (in steps of 90 percent, 75 percent, 50 percent, and 25 percent of rated power)
d. 2 hours at intermittent loads

4.2.5.2 Testing cycles. The tests shall be conducted in cycles, with each cycle comprising all of the above load points, and the cycle repeated until the specified durations of the tests are achieved.

4.2.5.3 Functional tests.

a. The engine starting and reversing appliances
b. The speed governor
c. The overspeed device
d. The lubricating oil failure indication device
e. The condition of turbocharger out of action (where applicable)
f. Running at idle speed

4.2.5.4 Component inspection. After all tests are completed, all main parts of the engine shall be dismantled and examined (see 6.2).

4.2.5.5 Measurements and recordings. The following particulars shall be measured and recorded:

a. Ambient test conditions:
    (1) Air temperature
    (2) Barometric pressure
    (3) Relative humidity
    (4) External cooling water temperature
    (5) Characteristics of fuel and lubricating oil

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b. Engine measurements:

(1) Engine power

(2) Engine RPM

(3) Brake torque

(4) Maximum combustion pressure, if available

(5) Exhaust smoke opacity

(6) Lubricating oil pressure and temperature

(7) Cooling water pressure and temperature

(8) Exhaust gas temperature in exhaust manifold and, if possible, at each cylinder outlet

(9) Turbocharger RPM

(10) Charge air pressure and temperature at the inlet and outlet of turbocharger compressor and charge air cooler

(11) Exhaust gas pressure and temperature at the inlet and outlet of turbocharger exhaust gas turbine

(12) Cooling water temperature at charge air cooler inlet and outlet

c. Results of examination done after type tests should include disassembling and examination of the main parts and the parts subject to wear.

4.2.5.6 Additional tests. For engines intended for different purposes and having different performances for each purpose, the type test program shall be extended to cover each performance under consideration of the most severe condition.

4.2.6 Endurance testing.

4.2.6.1 Test profile. The complete unit, including the engine, instruments, controls, and attached accessories, shall be subjected to a 1,500-hour endurance test using standard 8-hour endurance cycles (see table VI) unless otherwise approved by the Technical Authority (see 6.3.5). The target speed conditions shall be a percentage of the rated speed specified in table VI. The target load conditions shall be a percentage of the maximum load at the speed specified for that operating condition. When the engine is operated at 100-percent load or 110-percent load, the target load shall be greater than or equal to the rated BHP as specified (see 6.2). The engine idle speed shall be the normal minimum operating speed specified for the engine. For sequentially turbocharged engines, the endurance test cycle may be modified with approval by the Technical Authority. The standard endurance test shall consist of the steps given below and performed in the order given.

a. Measure gaseous and particulate exhaust emissions and exhaust smoke emissions in accordance with the specified exhaust emission requirements.

b. Conduct 8-hour endurance cycles in accordance with table VI.

TABLE VI. Standard 8-hour endurance cycle for main propulsion engines and diesel engine-generator sets. (1/, 2/, and 4/)

Step	Load 3/ (percent)	Speed (percent)	Time (minutes)
1	100	100	120
2	60	100	60
3	0	Idle	10
4	100	100	110
5	0	Idle	10
6	50	75	30

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TABLE VI. Standard 8-hour endurance cycle for main propulsion engines and diesel engine-generator sets – Continued. (1/, 2/, and 4/)

Step	Load 3/ (percent)	Speed (percent)	Time (minutes)
7	0	Idle	10
8	85	100	10
9 5/	110	100	120

NOTES:

1/ An alternative test profile may be required for main propulsion engines using sequential turbocharger systems or other special equipment. This shall be detailed in the first article test plan.

2/ Ship service generator set standard 8-hour endurance cycle is the same except that the speed shall be 100 percent of rated speed for all load points.

3/ For auxiliary service set, the standard endurance cycle shall be modified to suit the operating profile of the engine application such as hydraulic power units, pumps, etc.

4/ Load changes between starting to step 1 and step 1 to step 2 shall be accomplished within the required time listed in table IV.

5/ Upon completion of step 9, the engine shall be immediately shut down for a minimum of 10 minutes before beginning the next cycle.

4.2.6.2 JP-5 testing. When specified (see 6.2), 12 cycles (96 hours) of engine operation, as specified in table VI, shall be conducted using JP-5 (NATO F-44) fuel in accordance with MIL-DTL-5624. This may be considered as part of the endurance test. The 110 percent load point shall be the maximum achievable but not more than 110 percent. For sequentially turbocharged engines, the endurance test cycle may be modified with approval from the Technical Authority.

4.2.6.3 Endurance tests for engines with increased power ratings. Engines proposed under this condition shall be subjected to a 1,500-hour endurance test unless otherwise approved by the Technical Authority (see 6.3.5).

4.2.6.4 Periodic shutdowns. Each 8-hour cycle shall be followed by a minimum duration shutdown period of 10 minutes. The 10-minute shutdown time between cycles shall not be counted as part of the test qualification time as specified in table VI. The time to change from one load point of the test profile to another shall not exceed 20 seconds.

4.2.6.5 Engine load. For each engine operating condition, with the exception of the idle condition, the load shall be maintained within +1 percent or -0 percent of the target load value, and the speed shall be maintained within ±1 percent or ±10 RPM (whichever is less) of the target speed value.

4.2.6.6 Test fluids.

4.2.6.6.1 Fuel oil analysis. One sample of fuel used for test engine operation shall be collected at initial fueling and then at 250-hour intervals during the endurance test. Samples shall be labeled indicating test cycle number and endurance test hours. One sample of fuel from each lot of fuel shall also be collected. The fuel analysis shall determine the fuel's characteristics in accordance with MIL-DTL-16884 (or MIL-DTL-5624, if JP-5).

4.2.6.6.2 Lubricating oil consumption. Lubricating oil consumption shall be measured throughout the standard endurance test. An oil consumption test rig shall be connected to the engine lubricating oil system, and oil quantity shall be measured during the first hour of every cycle. Alternative oil consumption measuring techniques will be considered. All additions or removals of lubricating oil from the engine shall be measured. Lubricating oil consumption rates shall be determined at approximately every 250 hours of testing or as recommended by the manufacturer.

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4.2.6.6.3 Lubricating oil analysis. One lube oil sample shall be taken from each batch of test lube oil and then each lube oil circulating system at the following times during the endurance test: prior to the start of the testing, during the first cycle, during the last cycle, after approximately every 100 hours of testing, and during the cycle preceding and after every lube oil change. Physical and spectrographic analysis shall be conducted on each lube oil sample in accordance with the requirements of S9086-H7-STM-010/262.

4.2.6.6.3.1 Lubricating oil samples. With each sample sent to the laboratory, a fresh sample of unused lube oil from the same batch shall also be sent for control purposes. The sample shall be analyzed and compared with the fresh sample.

4.2.6.6.4 Jacket water analysis. One sample of jacket water shall be collected from the cooling system at the beginning and end of the endurance test and every 250 hours during the endurance test. A sample shall also be collected before and after any addition of inhibitor, before and after any change of jacket water. Samples shall be labeled. The sample shall be analyzed to determine inhibitor concentration.

4.2.6.7 Alternative test profile. An alternative test profile may be required for main propulsion engines to evaluate the engine's capability for load points that are unattainable or more stringent than those listed in table VI for some design configurations. A test profile shall be developed to evaluate the engine capability at the load points of interest. The alternative test profile shall be subject to approval by the Technical Authority (see 6.2).

4.2.6.8 Forced shutdowns. Forced shutdown(s) (see 4.2.3.10.3) that cause cessation of the test will require the test to be repeated starting from the first cycle.

4.2.6.9 Endurance test profile completion criteria. The test profile will be considered complete after the engine has successfully run all the conditions contained in the test profile, and the following conditions are met:

a. No shutdown or failure to start has occurred because of engine malfunction.

b. No forced shutdown has occurred during the test (see 4.2.3.10.3).

c. No internal component of the engine (i.e., parts requiring access cover removal) has been replaced, repaired, or adjusted during the test unless the component has been replaced as part of scheduled preventive maintenance in accordance with 4.2.3.9.

d. No external component of the engine has been replaced, repaired, or adjusted during the test unless the component has been replaced as part of scheduled preventive maintenance in accordance with 4.2.3.9.

e. Vibration levels shall not exceed the manufacturer's specifications. These specifications shall be provided prior to the start of testing.

f. Lubricating oil and jacket water analyses shall not exceed the manufacturer's trending limits for in-service engines.

g. Components presented for inspection shall not exceed the manufacturer's specifications. Engine manufacturer's specifications shall be provided prior to the start of testing.

h. All engine performance parameters are within the engine manufacturer's specifications and fuel consumption has not increased. Fuel consumption shall be within the manufacturer's specified limits. Performance parameters and fuel consumption specifications shall be provided prior to the start of testing.

i. The cumulative time due to unplanned engine shutdowns shall not exceed 1 percent of the test length. If the cumulative shutdown time exceeds the required 1 percent, the test may be continued as needed to achieve the 1 percent requirement. The shutdown period following each cycle, shutdowns due to facility problems, and shutdowns at the end of the test day do not count in determining the total unplanned engine shutdown time.

4.2.6.10 Inspections after completion of endurance test profile.

4.2.6.10.1 Visual and dimensional evaluation. Upon completion of endurance testing, the engine running clearances shall be measured. The engine and its attached and unattached assemblies shall be disassembled. Each piece shall be re-inspected and measurements made before commencing any other testing. Special tools furnished with the engine shall be used during this inspection. Additional inspections may be requested such as further disassembly of a particular component or nondestructive tests such as hydrostatic or crack detection tests.

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4.2.6.10.2 Reassembly. After the inspection is completed, the engine shall be reassembled to meet new product specifications and shall be subjected to the production unit inspection and tests including one cycle of the standard 8-hour endurance cycle.

4.2.6.10.3 Evaluation criteria. The failure of any component to meet the manufacturer's wear limits or the evidence of cracks, cavitation, pitting, scuffing, galling, fretting, or overheating shall be cause for the rejection of the engine. The wear limits are to be proposed by the manufacturer based on the test duration and agreed upon by the Technical Authority prior to the start of the test. Failure to meet evaluation criteria shall be cause for the rejection of the engine. The engine failure to meet the requirements of the specification shall result in a restart of the first article tests.

4.2.6.10.4 Special tools. Special tools furnished with the engines shall be validated on the engine during the post-trial examination.

4.2.6.11 Maintainability. Maintainability requirements shall be demonstrated during the post-first article test and examination.

4.2.6.12 Endurance test acceptance criteria. The approval test will be considered successful when the following conditions are satisfied:

a. All test profile completion criteria above are met

b. Any additional testing resulting from any forced shutdown, component replacement, or abnormal condition has been successfully completed

c. All lube oil and jacket water analyses are acceptable to the Technical Authority

d. Engine condition as determined by the inspection following the test profile is acceptable to the Technical Authority

e. Modifications developed as a result of testing or final engine inspections are proven effective and accepted by the Technical Authority

f. The cumulative time of unplanned engine shutdowns shall not exceed 1 percent of the test length

4.2.7 Other tests and examinations.

4.2.7.1 Emergency shutdown. This device shall be operated once, preceding the endurance test, without the engine running. Upon completion of the endurance test and before the post-endurance test examination, this device shall be actuated twice with the engine operating at no load and at a speed that is 10 percent greater than rated speed and twice with the engine operating at rated speed and full load. The engine shall be at full operating temperatures before each actuation of the shutdown device. The increase in engine speed prior to activating the shutdown device shall be continuous and the shutdown device shall be activated immediately upon reaching the required speed. For data collection purposes, a continuous measurement and record of the engine speed utilizing a strip recorder or equivalent shall be made for 1 minute before and 1 minute following each actuation.

4.2.7.2 Engine starting. The tests specified in 4.2.7.2.1 through 4.2.7.2.4 shall be conducted.

4.2.7.2.1 Normal start. The engine shall be subjected to 10 consecutive starts at ambient conditions with room temperature engine inlet air. The time to start, time to reach minimum idle speed, minimum starting RPM, minimum starting air pressure, and minimum total volume of air required for the start shall be determined during these start tests. If electric start, then a starting amperage profile shall be recorded. Critical lubricating oil pressures shall be monitored during the starting test. The time to start shall be the time from initial actuation of the starting mechanism with the engine at a standstill, to disengagement of the starting mechanism after which engine operation shall be sustained for a minimum duration of 30 seconds. The minimum starting RPM shall be the engine RPM at the time of starter disengagement after which engine operation continues for a minimum of 30 seconds. The minimum air pressure and total volume of air at that air pressure required to start the engine shall be measured by incrementally increasing the pressure of air supplied to the starting system. The time to start shall not exceed 5 seconds.

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4.2.7.2.1.1 Air starting test data. During the normal start tests, the following data shall be recorded:

a. The minimum quantity of air, expressed in cubic feet of free air at gage pressure of 0.0043 kg/m^2 (14.7 lb/in^2) and 20 °C (68 °F), necessary for each successive start.

b. The minimum air starting pressure at which the engine will start within the 5-second limit.

4.2.7.2.2 Cold start. The diesel engine shall be subjected to one cold start. The procedure shall be the same as normal start, except for the requirement to test at cold condition. Before conducting the cold start test, the engine shall be shut down for at least 12 hours. Ambient air temperature shall not exceed 1.7 °C (35 °F). Cold water at a temperature not greater than 1.7 °C (35 °F) shall be circulated through the engines until all parts of the engine are cooled to a reasonably uniform temperature. If cold starting aids are required, this shall be recorded. The time to start shall not exceed 10 seconds.

4.2.7.2.3 Emergency starting. When specified (see 6.2), the diesel engine shall be subjected to five emergency starts per the following:

a. Before the test starts, the engine shall be not run for 12 hours to ensure that it is at ambient conditions prior to the start of the test.

b. Each start shall be continued until reaching the rated load or the operating cycle of the starter is exceeded. The emergency start is successful if the generator is brought online and provides a full rated load within 60 seconds.

c. No starting aids or additional power sources outside of normal start sequence shall be supplied during the emergency start.

4.2.7.2.4 Submerged starting. When specified (see 6.2), the diesel engine shall be subjected to five submerged starts per the following:

a. To simulate shipboard conditions, a specified full load back pressure shall be set as the normal condition. The exhaust backpressure shall be recorded for each start.

b. The engine shall be started with the exhaust line closed until the back pressure in the exhaust line reaches a specified pressure and the exhaust line shall then be opened to reduce the back pressure to the normal condition that was preset to max backpressure at 100 percent load.

c. This test shall be repeated 4 times and is considered successful when the engine reaches rated speed. Data for volume of air or starter current must be recorded for each start after each start.

4.2.7.3 Governor response. The engine shall be tested to ensure compliance with the governing system response requirements (see 6.2). Engine overshoot shall be determined by taking the engine from low idle to high idle as rapidly as possible and then from high idle to low idle as rapidly as possible. With the dynamometer disconnected, the engine shall be operated at idle for 1-minute increments at every 100 RPM, as applicable to medium and high-speed engines, from low idle to high idle.

4.2.7.4 Piping integrity. The maximum working pressures and maximum leakdown rates of the various fluid piping systems external to the engine block shall be determined. These systems shall then be tested at 150 percent of their maximum working pressures. The fuel piping on the high-pressure side of a jerk-type fuel pump and high pressure common rail (HPCR) systems shall be excluded from this test. In lieu of pressure testing high-pressure fuel lines, all external high-pressure fuel delivery lines between high-pressure pumps and injectors may be protected with a jacketed piping system capable of containing fuel from a high-pressure line failure in accordance with SOLAS Chapter II-2, Regulation 15.2.9; the jacketed piping system shall include a means for collection of leakages, and arrangements shall be provided for an alarm in the event of a fuel line failure. To the extent possible, these pressure tests shall be conducted after all engine systems are fully assembled on the engine test stand. Those sections of piping that cannot be tested on the assembled engine shall be tested separately with each entrance and exit blanked off. These tests shall be carried out at room temperature using the fluids (e.g., lubricating oil, fuel oil, jacket water, seawater, or air) that are contained in the various sections of piping during engine operation. For each fluid piping system, the leakdown rate shall be determined by measuring the pressure at the start of the test and after 1 hour.

4.2.7.5 Waste heat. Testing shall be conducted to determine the waste heat available from the engine jacket water. The tests shall be run at standard operating conditions except that 1.7 °C (35 °F) maximum seawater shall be circulated through the cooler. If seawater is not readily available, then the test may be run with freshwater. The

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heat recovered from the jacket water while maintaining the minimum jacket-water temperature and the heat rejected to the seawater shall be measured at 25-, 50-, 75-, and 100-percent engine load.

4.2.7.6 Torsional vibration. The engine shall be instrumented and tested in accordance with MIL-STD-167-2, with the exception of "Torque across gears", "Simulated installation tests", and "Auxiliary branches". Testing shall include the operating points given in the operating profile. If it is impractical to conduct testing during the first article test, testing shall be conducted during shipboard trial tests.

4.2.7.7 Airborne noise. The engine shall be demonstrated to be in accordance with MIL-STD-1474 for airborne noise.

4.2.7.8 Structureborne noise. The engine shall be demonstrated to be in accordance with MIL-STD-740-2 for structureborne noise.

4.2.7.9 Engine alignment. Engine-to-dynamometer or engine-to-generator radial and facial alignment limits shall be determined. Engine-to-dynamometer, engine-to-generator, and crankshaft alignment shall then be measured before and after the endurance test. Crankshaft deflections shall be measured and recorded after the alignment.

4.2.7.10 Compression and firing pressure. Where test connections are installed or provided, firing pressures shall be taken for each cylinder at the beginning and end of the endurance test. Variables that the manufacturer's standard practice specifies to be controlled shall be measured during each test. Firing pressures shall be taken at rated load and speed. If the cylinder heads are not equipped with test connections, an instrumented cylinder head (special) shall be used on one cylinder and firing pressures shall be measured at the beginning and end of the endurance test. Cylinder leakdown tests, blow-by tests, or other tests recommended by the manufacturer to diagnose power train condition may be substituted subject to approval by the Technical Authority (see 6.2).

4.2.7.11 Fuel consumption. Fuel mass flow measurements shall be made at 100-, 90-, 70-, 50-, and 30-percent rated speed and each of the following loads at each speed: 100-, 80-, 60-, 40-, 20-, and 0-percent full corrected load (full fuel at the respective speed) when operating on F-76 fuel. The following conditions apply:

a. At any test speed where the test load would exceed the specified maximum load, the maximum load shall be substituted for those loads. At each engine operating condition, the engine shall be operated for at least 15 minutes. Once engine temperatures and fuel flow have stabilized at each engine operating condition, three fuel mass flow and other measurements shall be made. The test results shall be used to produce a fuel map.

b. For ship service generators and propulsion engines operating at constant speed, the load point shall be the same except that the speed shall be 100 percent of rated speed at all load points.

c. Fuel oil samples shall be taken prior to the fuel consumption test and fuel density shall be reported in 4.2.6.6.1.

4.2.7.12 Fuel performance comparison. Fuel consumption shall also be measured using JP-5 and compared to the results obtained in 4.2.7.11. Fuel mass flow measurements shall be made using Aviation Turbine Fuel, grade JP-5 (F-44) (in accordance with MIL-DTL-5624), at 100-, 90-, 70-, 50-, and 30-percent rated speed and each of the following loads at each speed: 100-, 80-, 60-, 40-, 20-, and 0-percent full corrected load (full fuel at the respective speed). The following conditions apply:

a. At any test speed where the test load would exceed the specified maximum load, the maximum load shall be substituted for those loads. At each engine operating condition, the engine shall be operated for at least 15 minutes. Once engine temperatures and fuel flow have stabilized at each engine operating condition, three sets of fuel mass flow and other measurements shall be made.

b. For ship service generator and propulsion engines operating at constant speed, the load point shall be the same except that the speed shall be 100 percent of rated speed at all load points.

c. JP-5 samples shall be taken prior to the fuel consumption test and fuel density shall be reported.

4.2.7.13 Overspeed governor. With the engine governor deactivated or set at an RPM greater than the overspeed protection device activation RPM as recommended by the engine manufacturer and agreed upon by the Technical Authority, the engine speed shall be increased from idle to the RPM at which the overspeed protection device is activated. This RPM traverse shall be completed between 30 seconds and 1 minute. Continuous measurements of RPM shall be taken during the RPM traverse and for 30 seconds after the overspeed protection device is activated.

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4.2.7.14 Exhaust emission. Gaseous and particulate exhaust emissions (NO_X, CO, carbon dioxide [CO_2], HC, oxygen [O_2]) shall be measured by utilizing the methods outlined in ISO 8178-1 (see 6.2). The cycle used for the test shall conform to ISO 8178-4. Test cycle "E3" shall be used for the main propulsion engine and test cycle "D2" for generator engines. Gaseous and particulate exhaust emissions shall be measured and confirmed at the beginning and at the end of the standard endurance test. Additionally, exhaust smoke emissions shall be measured during the first and last standard 8-hour endurance test cycle.

4.2.7.15 Alarms and shutdowns. Each of the applicable alarms and shutdowns as listed in 3.7.9 shall be tested. The following conditions apply:

a. The parameter activating each switch shall be increased or decreased from the acceptable operating range to the alarm activating point. Static tests are acceptable in lieu of testing alarm switches during engine operation. The alarm activation value of each parameter shall be measured for each test.

b. Standard commercial test cells, detached accessories, and instrumentation may be used for these tests.

c. A failure during the test cycle shall be cause for rejection.

4.2.7.16 Turbocharger destructive testing. This requirement shall be demonstrated by actual destructive testing or by engineering analysis of this capability by the turbocharger manufacturer.

4.2.7.17 Torsiograph. When required (see 6.2), torsiograph tests shall be conducted in accordance with type III of MIL-STD-167-2.

4.2.7.18 Shock testing. When specified (see 6.2), shock tests shall be conducted in accordance with the requirement for grade A, hull mounted requirements of MIL-DTL-901 and as specified herein. The complete unit including the engine, marine gear (if attached directly to the engine), generator, instruments, controls, and attached and detached accessories shall be tested as a principal unit using the heavyweight shock machine specified in MIL-DTL-901. Assemblies and subassemblies furnished by the manufacturer in accordance with this specification shall be production components. Other items not furnished under this specification shall be in accordance with the approved shock test procedure (see 6.2).

4.2.7.18.1 Examinations before shock test. Before starting the shock test, the unit shall be examined and the following instrumentation evaluation, positions marking, and component measurement shall be made:

a. Velocity pickups, instruments for measuring unit operation, and provisions for remote shutdown shall be inspected.

b. Scribe lines or punch marks to indicate the exact location of attached accessories, components, and the major housing to detect any movement during the test.

c. Measure crankshaft web deflection for all throws, and crankshaft thrust clearances.

d. Draw a lubricating oil sample and perform an analysis in accordance with the first article test procedure (see 4.2.6.6.3).

4.2.7.18.2 Pre-test evaluation criteria. Any flaws or undesirable conditions that might contribute to shock damage or be mistaken for shock damage during the examination after the shock test shall be cause for the rejection of the unit.

4.2.7.18.3 Examinations after shock test. After completion of the shock tests, and before removal from the shock platform, the engine and its accessories shall be inspected in accordance with MIL-DTL-901, and as follows:

4.2.7.18.3.1 Preliminary evaluation. After completion of the shock tests and before removal from the shock platform, the engine and its accessories shall be inspected as follows:

a. Visually examine the complete unit for breakage, distortion, cracks, leaks, and relative movement between components, housings, major assemblies, and accessories.

b. Remove inspection covers and visually examine accessible internal areas for damage.

c. Measure and record the crankshaft deflection for all throws, and crankshaft thrust clearances.

d. Draw a lubricating oil sample and analyze in accordance with the first article inspection.

e. Inspect all lubricating oil filters and strainers.

f. Remove one main bearing and one connecting rod bearing for inspection.

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4.2.7.18.3.2 Detailed examination. The engine unit shall be returned to the manufacturer or to a specified Navy facility for more detailed examinations and operational tests. Upon receipt at the final inspection site and before removal from the carrier, the unit shall be given a visual inspection to ensure that no damage occurred during transit. At the final inspection site, the unit shall be inspected as follows:

a. Visually examine the complete unit for breakage, distortion, cracks, leaks, and relative movement between components, housings, major assemblies, and accessories.

b. Remove inspection covers and visually examine accessible internal areas for damage.

c. Measure and record the crankshaft deflection for all throws, and crankshaft thrust clearances.

d. Conduct hydrostatic tests of piping systems and coolers at 150 percent of the maximum working pressures.

4.2.7.18.4 Shock test evaluation criteria. The failure of any component to meet the manufacturer's wear limits or the evidence of cracks, cavitation pitting, or signs of stress or overheating shall be cause for rejecting the engine. The unit shall exhibit none of the impermissible damage specified in 4.2.7.18.4.1.

4.2.7.18.4.1 Impermissible damage. Damage as characterized or as specified as follows shall not be permitted as a result of the shock test.

a. Damage that will result in failure to maintain engine operation within the limits of the manufacturer's operating parameters, rated power, or rated speed or failure to maintain lubricating oil from exceeding the manufacturer's condemning limits.

b. Damage that causes or necessitates reduction of equipment output to a level below rated for any length of time within 30 days after an attack.

c. Damage that renders inoperative any item or device installed to protect the equipment or operating personnel, such as speed limiting governors, over-speed trips, high-pressure relief valves, low-lube oil pressure alarms, or any other shutdown devices.

d. Shock-induced actuation of safety devices specified in 4.2.7.18.4.1c.

e. Damage or shock-induced mis-adjustment of speed governing systems that result in the inability to automatically maintain speed within the limits imposed by the individual equipment specifications until repairs or adjustments are made.

f. Loosening or shifting of any item affixed to or housing a rotating part.

g. Damage that causes any assembly or subassembly to detach from any of its intended restraints.

h. Damage that requires repairs of any type to be made before equipment operating parameters can be adjusted to suit operational requirements.

i. Damage that creates a safety hazard to personnel or equipment.

j. Complete fracture of any unit or welded assembly, including any supporting chock material.

k. Stripping or failure by gross distortion of all engaged threads (male or female) of any threaded part or fracture (cracking) of any fastener.

l. Increase of dynamic imbalance of any rotating part or assembly to such an extent that the affected item requires rebalancing to meet the specified balance tolerance.

m. Damage causing misalignment, requiring correction before the engine can be operated at rated speed and power.

4.2.7.18.4.2 Permissible damage. Damage from shock tests as characterized or specified herein will be permitted to the extent or under the conditions listed for each item.

a. Damage that shortens the useful life of wearing parts by initiating or aggravating some long-term detrimental effect such as wear, fatigue, or erosion is permissible if the useful life of wearing parts is reduced by no more than 25 percent.

NOTE: Due to difficulties associated with estimating reduction of useful life, equipment acceptability shall be determined with consideration for the accuracy of the estimate. In cases where it is especially difficult to estimate percent reduction of useful life, equipment will not be rejected for this cause unless the limits specified herein have been grossly exceeded. In cases where the availability of test or maintenance information permits a more accurate

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estimate of useful life reduction to be made, appropriately less leniency will be allowed when determining acceptability of the equipment on this basis.

b. Damage resulting in minor leaks of fluids or gases is permissible insofar as no portion of this specification is violated by any consequences of the leakage.

c. Damage that requires adjustment to control or equipment set points to regain proper operation or damage that requires repairs or compensatory action to prevent further (and eventually unacceptable) damage that is related to increased wear rates or other long-term effects.

d. Loosening of hold-down devices, if no requirement of this specification is violated as a consequence of such loosening.

e. Damage that results in continuous or intermittent rubbing contact between stationary and moving parts (during equipment operation) in areas where it was not originally intended that such contact be made is not permitted. On a case-by-case basis, the Technical Authority will adjudicate rubbing that quickly wears away without causing adverse effects that otherwise violate any portion of this specification.

f. Damage that brings about (until repairs are made) a requirement for slight to moderate increase of manual effort, maintenance, or attention on the part of equipment operators in order to maintain the operating mode or the machinery space environment.

4.2.7.18.5 Reassembly. After the detailed examination is complete, the engine shall be reassembled and subjected to post-shock tests in accordance with 4.2.7.18.6.

4.2.7.18.6 Post-shock tests. If the examinations after the shock test reveal no impermissible damage, and if the unit is capable of full power operation, the unit shall be subjected to the following post-shock tests. The attached and detached accessories that were previously with the unit during the shock test shall remain with the unit for the post-shock tests. During these tests, the instruments and controls shall be checked for correct operation and accuracy. Upon completion of the post-shock tests, a lubricating oil sample shall be obtained and analyzed in accordance with 4.2.6.6.3.

a. Four 8-hour endurance cycles of the appropriate type as specified in table VI with each cycle being followed by a 10-minute minimum duration shutdown period.

b. Startup tests in accordance with 4.2.7.2.

c. Test the emergency shutdown device to ensure it is fully operational.

d. Measure governor response and overspeed protection device to ensure both are fully operational.

e. Verify response of the following alarms at least three times: low lubricating oil pressure, high lubricating oil temperature from engine, high inlet manifold air temperature (turbocharged engines only), high water outlet temperature, high crankcase pressure or low vacuum, and high exhaust temperature. Static tests are acceptable in lieu of testing the alarm switches during engine operation.

4.2.7.18.7 Post-shock examination. After the post-shock testing specified in 4.2.7.18.6, the unit shall be disassembled, and bearings and moving parts shall be examined for excessive wear or damage attributed to the shock and post-shock tests. The internal surfaces of the block, housings, accessories, and mounting hardware shall be examined for evidence of shifting, rubbing, momentary contact of moving parts, and cracks.

4.2.7.18.8 Reassembly. After the post-shock examination is completed, the engine shall be reassembled.

4.2.7.18.9 Shock test acceptance criteria. The failure of any component to meet the manufacturer's wear limits or the evidence of cracks, cavitation pitting, or signs of stress or overheating shall be cause for rejecting the engine. The unit shall exhibit none of the impermissible damage specified in 4.2.7.18.4.1.

4.2.7.19 Low load testing. When specified (see 6.2), low load testing shall be conducted on the same engine as all other first article testing. The purpose of this testing is to demonstrate satisfactory engine operation at low load in the range of 15 to 50 percent of rated power.

4.2.7.19.1 Low load test and inspection. Table VII provides the low load test conditions required to establish and validate satisfactory engine operation. The engine may be subject to additional tests as determined by the Technical Authority (see 6.2). The test shall be based on the maximum ambient conditions listed in 3.3.1. Unless otherwise specified (see 6.2), the test shall be conducted using F-76 as specified in 4.2.3.6.

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TABLE VII. Standard low load test for main propulsion engines and diesel engine generator sets.

Step	Load [1/] (percent)	Speed [2/] (percent)	Time (hours)
1	100	100	2 (minimum)
2	15 [1/]	[2/]	96
3	Post-96-hour continuous run inspection (see 4.2.7.19.20).		
4a [3/]	50	[2/]	1
Shutdown for visual and borescope inspection to determine presence of oil or carbon accumulation.			
4b [4/]	60	[2/]	1
Shutdown for visual and borescope inspection to determine presence of oil or carbon accumulation.			
5	Post-test inspection (see 4.2.7.19.20).		
6	20	[2/]	1 (minimum)
7	30	[2/]	1 (minimum)
8	40	65	1 (minimum)
9	50	[2/]	1 (minimum)
10	Shutdown for visual and borescope inspection to determine presence of oil or carbon accumulation.		
11 [5/]	50	11	50
12	Post-test inspection (see 4.2.7.19.20).		

NOTES:

[1/] An alternative test profile shall be required for main propulsion engines to evaluate the engine's capability for low load points that are unattainable or more stringent than those listed in this table for other design configurations. When specified (see 4.2.7.19.17), a test profile will be developed to evaluate the engine capability at the low load points of interest.

[2/] Ship service generator set speeds shall be 100 percent of the rated speed for all load points. For propulsion engines, speed shall be the corresponding speed on the propeller curve at that load point.

[3/] After completion of Step 3, if there is oil or carbon accumulation, Step 4a shall be conducted. Start the engine and load to 50 percent load. Record exhaust opacity during the entire step.

[4/] After completion of Step 4a, if there is oil or carbon accumulation, Step 4b shall be conducted. If there is oil or carbon accumulation after Step 4b, a water wash of the turbocharger compressor (inlet) shall be performed prior to securing the engine, if applicable.

[5/] If there is oil or carbon accumulation after Step 11, Step 4b shall be conducted. If there is oil or carbon accumulation after Step 4b, a water wash of the turbocharger compressor (inlet) shall be performed prior to securing the engine, if applicable.

4.2.7.19.2 Engine material condition baseline. An engine material condition baseline shall be established prior to the start of testing that will support the post-testing inspection requirements. The material condition baseline shall include visual appearance of accessible wearing surfaces and a record of torques, thrust measurements, hydraulic pressures, and other measurement data to establish the assembled condition of the engine.

4.2.7.19.3 Acceptance criteria. Acceptance criteria for all portions of the material condition baseline shall be established prior to the start of testing.

4.2.7.19.4 Preventive maintenance. Preventive maintenance to be conducted during the test profile shall be approved prior to conducting the test. The approved preventive maintenance shall be the only allowed maintenance performed on the engine not subject to the test acceptance criteria.

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4.2.7.19.5 Inspections before low load testing. The first article inspection and test requirements in accordance with this document shall be satisfactorily completed prior to the commencement of the low load test.

4.2.7.19.5.1 Measurement standards. The requirements of 4.2.3.2 are applicable to the low load test.

4.2.7.19.5.2 Inspection results. The results of all the inspections conducted at the conclusion of the first article test in accordance with 4.2 shall be available prior to the start of the low load test. If the low load testing is not being conducted on the same engine that was subjected to the first article test, material baseline requirements may be evaluated (see 6.2).

4.2.7.19.6 Evaluation criteria. The requirements of 4.2.6.10 are applicable to the low load test. If a failure is identified in accordance with 4.2.6.10.3, no low load testing shall be conducted until a satisfactory first article test is completed.

4.2.7.19.7 Low load test operating conditions. The requirements of the first article test are applicable to the low load test.

4.2.7.19.8 Operation criteria. Upon activation and stabilization of the lubricating-oil preheat system, the jacket water keep-warm system, and the lubricating-oil priming system, the engine shall be started and run continuously through Steps 1 and 2 of table VI. The inspections in accordance with Step 3 of table VI shall immediately follow the completion of Step 2. Step 4 shall be performed immediately after the conclusion of the inspections of Step 3 and restoration of the engine to operating condition.

4.2.7.19.9 Commencing or recommencing. Upon activation and stabilization of the lubricating-oil preheat system, the jacket water keep-warm system, and the lubricating-oil priming system, the engine shall be started and run continuously through Steps 1 and 2 of table VI. The inspections per Step 3 of table VI shall immediately follow completion of Step 2. Step 4 shall be performed immediately after the conclusion of the inspections of Step 3 and restoration of the engine to operating condition.

4.2.7.19.10 Engine monitoring. Engine operation shall be monitored throughout the low load test. Specified parameters shall be measured every hour during the 96-hour low load test after the engine has been operating at a particular condition for at least 10 minutes and the temperature and pressures have stabilized. For the 50 percent of rated load test (table VI, Step 4), specified parameters shall be measured every 10 minutes. Test events shall be monitored. Charge air cooler performance shall be monitored and verified within limits. The high-temperature and low-temperature pre-heating system performance shall be within limits.

4.2.7.19.11 Engine modification. The requirements of 4.2.3.9 are applicable to the low load test.

4.2.7.19.12 Engine shutdowns. The requirements of 4.2.3.10 are applicable to the low load test.

4.2.7.19.13 Performance tests conducted before, during, and after the low load test.

4.2.7.19.13.1 Compression and firing pressure. The requirements of 4.2.7.10 are applicable for the low load test. If no test connections are provided, this test is not required.

4.2.7.19.13.2 Fuel mass flow and other measurements. Once engine temperatures and fuel flow have stabilized at each engine operating condition, three fuel mass flow and other measurements shall be made. These test results along with the results obtained in accordance with 4.2.7.11 shall be used to supplement the fuel map required in 4.2.7.12.

4.2.7.19.13.2.1 Engines operating at constant speed. For engines operating at constant speed, the load point shall be the same except that the speed shall be 100 percent of rated speed at all load points.

4.2.7.19.14 Exhaust smoke and opacity. Exhaust smoke and opacity shall be monitored continuously during the low load test. During load changes, engine temperatures and fuel flow shall be permitted to stabilize prior to measurement of the opacity, which shall be 20 percent or less. Continuous measurements shall be taken for all transient and non-steady-state conditions.

4.2.7.19.15 Lubricating oil analysis. The requirements of 4.2.6.6.3 are applicable for the low load test with following modification: two lube oil samples shall be taken prior to the start of the 96-hour low load test and at the conclusion of this test.

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4.2.7.19.16 Standard low load test profile.

4.2.7.19.16.1 Low load test profile. The complete unit, including the engine, instruments, controls, and attached accessories, shall be subjected to the test profile of table VII and as specified below. The target speed conditions shall be a percentage of the rated speed specified in table VII. The target load conditions shall be a percentage of the maximum load at the speed specified for that operating condition. The standard low load test steps shall be performed in the order given in table VII.

4.2.7.19.16.2 Engine operating conditions. For each engine operating condition, the load shall be maintained within plus 1 percent or minus 0 percent of the target load value, and the speed shall be maintained within ±1 percent or ±10 RPM (whichever is less) of the target speed value.

4.2.7.19.17 Alternative test profile. An alternative test profile will be required for main propulsion engines to evaluate the engine's capability for low load points that are unattainable or more stringent than those listed in table VII for other design configurations. A special test profile shall be developed to evaluate the engine capability at the low load points of interest. The alternative test profile shall be subject to approval by the Technical Authority (see 6.2).

4.2.7.19.18 Forced shutdown. Forced shutdown(s) (see 4.2.3.10.3) that cause cessation of the test shall require the test to be repeated starting from Step 1.

4.2.7.19.18.1 Notification of forced shutdown. An abnormal condition may be identified by either the manufacturer or the Government representative monitoring the testing. Supporting documentation, analyses, and rationale detailing the cause, corrective action, and impact on the test completion criteria shall be submitted to the Technical Authority for evaluation to determine if additional testing is warranted (see 6.3.7).

4.2.7.19.19 Low load test profile completion criteria. The low load test profile will be considered complete after the engine has successfully run all the conditions contained in the Low Load Test Profile and the following conditions are met:

a. No shutdown or failure to start has occurred because of engine malfunction.

b. No forced shutdown has occurred during the test as defined in 4.2.3.10.3.

c. No internal component of the engine (i.e., parts requiring access cover removal) has been replaced, repaired, or adjusted during the test unless the component has been replaced as part of scheduled preventive maintenance in accordance with 4.2.3.9.

d. No external component of the engine has been replaced, repaired, or adjusted during the test unless the component has been replaced as part of scheduled preventive maintenance in accordance with 4.2.3.9.

e. Linear vibration levels shall not have exceeded the manufacturer's limits for production acceptance. These limits shall have been specified prior to the start of testing.

f. Lube oil analyses shall not exceed the manufacturer's trending limits for in-service engines.

g. Components presented for inspection shall not have exceeded the manufacturer's wear limits if measurements are taken or appearance limits if comparison photographs are used for production acceptance.

h. All engine performance parameters are within production limits.

i. Measurement of the opacity (steady state) shall be 20 percent or less.

4.2.7.19.20 Inspections after low load tests.

4.2.7.19.20.1 Visual and dimensional evaluation.

a. After completion of low load test step 2 (96 hours at 15 percent), an inspection of the turbocharger or supercharger, engine combustion intakes, and exhaust areas shall be performed to check for accumulation of oil and soot upon completion of the 96-hour low load test. A light coating of oil on surfaces is acceptable. Standing or pooling of oil is not acceptable. A light layer of loose soot or carbon buildup is acceptable.

b. After completion of the low load test, an inspection of the turbocharger or supercharger, engine combustion intakes, and exhaust areas shall be performed to check for accumulation of oil and soot upon completion of the low load test. A light coating of oil on surfaces is acceptable. Standing or pooling of oil is not acceptable. A light layer of loose soot or carbon buildup is acceptable.

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4.2.7.19.20.2 Repetition of inspections. Upon completion of all testing, the inspections listed in 4.2.7.19.5 shall be repeated on the same components.

4.2.7.19.21 Reassembly. After the inspection is completed, the engine shall be reassembled to meet new product specifications and shall be subjected to the Production Unit Inspection and Tests including one cycle of the standard 8-hour endurance cycle.

4.2.7.19.22 Evaluation criteria. The failure of any component to meet the manufacturer's wear limits or the evidence of cracks, cavitation, pitting, scuffing, galling, fretting, or overheating shall be cause for the rejection of the engine.

4.2.7.19.22.1 Exhaust opacity failure. Failure to meet the exhaust opacity requirements shall be cause for the rejection of the engine. Unacceptable accumulation of oil and soot shall be cause for the rejection of the engine.

4.2.7.19.22.2 Evaluation criteria failure. Failure to meet evaluation criteria shall be cause for the rejection of the engine and shall result in a restart of the first article tests.

4.2.7.19.23 Test acceptance criteria. The low load test shall be considered successful when the following conditions are satisfied:

a. All test profile completion criteria in 4.2.6.9 are met

b. Any additional testing resulting from any forced shutdown, component replacement, or abnormal condition has been successfully completed

c. All lube oil analyses are acceptable

d. Engine condition, as determined by the inspection following the low load test profile, is acceptable

e. Modifications developed as a result of testing or final engine inspections are proven effective and accepted by the Technical Authority

f. Exhaust opacity is less than 20 percent during the 96-hour portion of the low load test. Exhaust opacity not to exceed 20 percent for more than 3 minutes during transients

g. No unacceptable accumulation of oil or soot found during the inspections

4.2.8 First article report. The first article report shall be prepared as specified (see 6.2 and 6.3.6).

4.3 Conformance inspection and testing. Conformance test and inspection shall be performed on a complete engine or engine package in accordance with table III. These inspection and test requirements are applicable for all production engine models following satisfactory completion of the first article inspection and tests for the first engine of the model. Each engine tested under this category shall be subjected to the minimum testing in accordance with the requirements of table III.

4.3.1 Main propulsion engines. Conformance testing of main propulsion engines shall consist of the requirements specified in table III and as described for first article testing with the following exceptions:

a. One standard 8-hour endurance cycle composed as specified in table VI.

(1) Monitored parameters shall include those items on the main propulsion engines specified in 4.2.3.8. Measurements of monitored parameters specified by the manufacturer shall be made at intervals specified in the engine monitoring requirements of the first article inspection test.

(2) Lubricating oil samples. Lubricating oil samples shall be collected before and after the cycle and analyzed as specified in the first article inspection.

(3) Exhaust smoke emission measurements. Exhaust smoke emission shall be measured using the methods specified for the first article inspection at each operating point of the standard 8-hour endurance cycle.

b. Response of each of the following alarm switches shall be tested three times. The parameter activating each switch shall be increased or decreased from the acceptable operating range to the alarm activating point. Static tests are acceptable in lieu of testing alarm switches during engine operation. The alarm activation value of each parameter shall be measured for each test.

(1) Low lubricating-oil pressure

(2) High lubricating-oil temperature from engine

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(3) High inlet manifold air temperature (for turbocharged or supercharged engines only)

(4) High water outlet temperature

(5) High crankcase pressure or low vacuum

(6) High exhaust temperature

c. Standard commercial test cells, detached accessories, and instrumentation may be used for these tests.

d. A failure during the test cycle shall be cause for rejection (see 4.2.6.10.3). A repaired engine shall be retested for one cycle as specified in table VI.

4.3.2 Generator set. Each generator set shall undergo the conformance inspection and tests in accordance with the requirements of table III and 4.2.4 except that the 8-hour test shall use the generator set 8-hour endurance cycle as specified in table VI, note 2/ and engine alignment shall be tested.

4.3.3 Auxiliary service sets. Each auxiliary service set such as hydraulic units and engine driven pumps shall undergo the conformance inspection and tests in accordance with the requirements above except that the 8-hour test shall use the auxiliary service unit test cycle specified in table VI, note 5/.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity. For delivery to NAVSUP or DLA, packaging must be in accordance with MIL-STD-2073-1.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The engines are intended for use in naval vessels as propulsion or powering auxiliary equipment. This specification covers two− or four−stroke engines; inline, vee, or opposed piston configurations; liquid cooled; and naturally aspirated, supercharged, or turbocharged. The engines will be used for both continuous and non-continuous duty. The diesel engines will vary in loading from high time at low load to high time at high load. Current stock of diesel engines and their components and parts may be used until depleted; future requisitions for items covered in this specification should refer to the current version of this specification. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained as approved by the Technical Authority.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1).

c. Whether the engine is procured alone, with accessories, or as a complete package with enclosure (see 3.1).

d. If the engine installation will be other than parallel to the centerline of the ship (see 3.4.1).

e. When shock resistance is required (see 3.4.2).

f. Details required for a recommended spares list to support 90 days of shipboard operation (see 3.5.1 and 6.7).

g. When an MEA is required (see 3.5.2).

h. When additional accessory equipment must be furnished, if other than as specified (see 3.6.1).

i. When other accessories or engine attached piping necessary to facilitate engine operation is required, if other than specified (see 3.6.1).

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j. When piping required to connect detached accessories, instruments, and controls to the engine, or to cooling water, fuel starting air, electrical, exhaust, or lubrication system must be furnished, if other than specified (see 3.6.1).

k. Whether guards, for flywheels and other exposed moving parts, to protect personnel will be supplied by the manufacturer for installation by the installing activity (see 3.6.2.1).

l. Minimum fuel viscosity and whether a cooler will be necessary (see 3.7.3.5).

m. Whether combustion air will be taken from the engine compartment or from ducts provided by the installing activity (see 3.7.5).

n. Noise requirements (see 3.7.5.3 and 3.7.6).

o. Whether 40 CFR or MARPOL Annex VI emissions requirements apply (see 3.7.6.2).

p. Whether emergency start testing is required (see 3.7.7 and 4.2.7.2.3).

q. Whether submerged start testing is required (see 3.7.7 and 4.2.7.2.4).

r. Whether starting batteries are required (see 3.7.7.3).

s. Operating controls, if other than as specified (see 3.7.10).

t. Whether a low load test will be performed (see 3.8 and 4.2.7.19).

u. Whether accessory drawings are to be provided (see 3.11 and 6.4).

v. Whether manuals are to be provided (see 3.12 and 6.5).

w. Whether a Government Witness is required (see 4.2.2).

x. Whether measurement standards must be submitted for approval by the Technical Authority (see 4.2.3.2).

y. Whether or not first article testing must be conducted under operating conditions other than specified (see 4.2.3.4).

z. Request to use alternative fuels other than F-76 or JP-5 and for which tests, must be submitted for approval by the Technical Authority (see 4.2.3.6).

aa. Request to deviate from table IV (ramp time) must be submitted for approval by the Technical Authority (see 4.2.3.7).

bb. Details required for first article test procedure (see 4.2.3.10.1, 4.2.3.10.2, and 4.2.7.18).

cc. Details required for first article test plan (see 4.2.3.12).

dd. Details on power correction method (see 4.2.4.3).

ee. Component inspection details (see 4.2.5.4).

ff. Rated BHP when the engine is operated at 100 percent load or 110 percent load (see 4.2.6.1).

gg. Uprating requirements (see table VIII, note 1/).

hh. When a commercial engine with documented service records is proposed, the documented records must be submitted for review to the Technical Authority to determine test requirements (see table VIII, note 2/).

ii. When design or manufacturing changes are made to a previously qualified engine or to a qualified engine currently in production, copies of the revised drawings must be submitted for review as part of the design assessment, where the engine manufacturer may submit their own Engineering Change Proposal (ECP) indicating drawing number(s), stating the nature of the change, along with a statement indicating the design change has been tested or submitting calculations, and stating the design change has not affected engine performance or has resulted in an improvement to the old part, and it must also be signed by the person in charge of engineering (see table VIII, note 3/).

jj. For other U.S. Government qualified engines, the test specification the engine was certified to, test reports, and documented service record must be submitted for review to determine if test requirements can be modified (see table VIII, note 4/).

kk. When an engine is proposed that contains significant changes that affect the engine components, systems, or changes that affect engine performance, shaft dynamics, or cycle temperatures, including major design changes or minor modifications made to improve the manufacturing process or service life, or any component modified such

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that it is no longer interchangeable with the component it replaces, information must be submitted for review to the Technical Authority to determine test requirements (see table VIII, note 5/).

ll. Whether 96 hours (12 cycles) of engine operation on JP-5 is required (see 4.2.6.2).

mm. Request to use alternative test profile must be submitted for approval by the Technical Authority (see 4.2.6.7).

nn. Governing system test requirements (see 4.2.7.3).

oo. Request to use alternative means to diagnose compression and firing pressures must be submitted for approval by the Technical Authority (see 4.2.7.10).

pp. Emissions requirements, if different from what is specified (see 4.2.7.14).

qq. Whether torsiograph tests must be performed (see 4.2.7.17).

rr. Whether shock testing must be performed (see 4.2.7.18).

ss. Whether the engine will be subject to additional low load tests (see 4.2.7.19.1).

tt. Whether the low load test will be conducted with a fuel other than F-76 (see 4.2.7.19.1).

uu. Whether material baseline requirements need to be evaluated (see 4.2.7.19.5.2).

vv. Request to use alternative test profile must be submitted for approval by the Technical Authority (see 4.2.7.19.17).

ww. Details required and approval authority for first article test report (see 4.2.3.11 and 4.2.8).

xx. Packaging requirements (see 5.1).

yy. Which documentation is recommended to be requested (see 6.3).

6.3 Recommended documentation. The following documentation is recommended to be requested as part of the acquisition (see 6.2) of any diesel engine under this specification. All documentation should be submitted for evaluation and approval by the Technical Authority.

6.3.1 Dissimilar metals. Documentation identifying contact between dissimilar materials, if used, and the means to minimize galvanic corrosion, means of corrosion protection for aluminum parts, and the internal and external coatings in the engine.

6.3.2 Torsional vibration. Documentation of the analysis conducted for torsional vibration.

6.3.3 Dimensions and weight. Dimensions should include projections of attached accessories, levers, and piping. Weight should include the maximum weight of the engine in operating condition and include the weight of attached and detached accessories and service fluids.

6.3.4 Emissions. Technical documentation for onboard verification and records demonstrating that the engines have been tested by the manufacturer at the proposed rating for gaseous and particulate emissions (NO_X, CO, CO_2, HC, O_2, and PM) and smoke emissions (opacity).

6.3.5 Reduced requirements for first article testing. For certain engine designs with documented service history, the Technical Authority may reduce the first article test and inspection requirements, or waive them entirely. Table VIII provides the design and service history categories for which the Technical Authority will consider a reduction or waiver of testing. Bidders offering such products, who wish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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TABLE VIII. Endurance test hours for first article testing of various engine categories.

Engine category based on design and service history	Endurance test hours and [number of cycles] for 100% rating	Endurance test hours and [number of cycles] for 1-10% uprating 1/	Endurance test hours and [number of cycles] beyond 10% uprating 1/
R & D development or commercial engine with no documented service record	1,500 [188]	N/A	N/A
Commercial engine with documented service record 2/	500 [63]	500 [63]	1,000 [125]
Other US Government qualified engines (test and documented service record) 3/, 4/	500 [63]	500 [63]	1,000 [125]
Navy qualified engines (see 6.6.4)/engines with Navy machinery certificate for naval applications with significant changes 3/, 5/	250 [31]	500 [63]	1,000 [125]
Navy qualified engines/engines with Navy machinery certificate for naval applications with no significant changes with no documented service record 3/, 5/	No testing required 6/	500 [63]	1,000 [125]
Navy qualified engines/engines with Navy machinery certificate for naval applications with no significant changes with documented service record 3/, 5/	No testing required 6/	No testing required 6/	1,000 [125]

NOTES:

1/ The required number of hours of performance testing to validate 100 percent rating must be satisfactorily completed prior to conducting the uprating test (see 6.2).

2/ When a commercial engine with documented service records is proposed, the documented service records should be submitted that contain cumulative service of at least 25,000 running hours in marine or industrial installations using liquid fuels on engines with a rated power equal to or greater than the proposed rating for review by the Technical Authority to determine test requirements (see 6.2). At least two of these engines must have a minimum of 5,000 running hours each with no overhauls except published scheduled maintenance. Consideration will be given to diesel engines of identical make, model, engine family, and cylinder rating, but with different number of cylinders if satisfactory service can be demonstrated.

3/ When design and manufacturing changes are made to a previously qualified engine or to a qualified engine currently in production, information must be provided to the Technical Authority as specified (see 6.2) to determine test requirements.

4/ For other U.S. Government qualified engines, the test requirements may be modified by the Technical Authority when specified (see 6.2).

5/ Significant changes are defined as modifications made to the engine components, systems, or changes that affect engine performance, shaft dynamics, and cycle temperatures. The modifications may be major design changes or minor modifications to improve manufacturing processes or service life. Any engine component modified such that it is no longer interchangeable with the component it replaces is considered to have undergone a significant change and information must be provided to the Technical Authority to determine test requirements (see 6.2).

6/ For diesel generator applications, any previous Navy qualified engine that does not require any testing as indicated in the table is to undergo a packaging test in accordance with 4.2.4.2.

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6.3.6 First article report. Test report that includes all data collected during the testing required in Section 4, in addition to the following:

a. Any discovered design deficiency (or evidence of design discrepancies), including an analysis of the discrepancy.

b. A maintainability demonstration report.

c. Visual documentation (via photography or videography) of the various engine components and engine areas including piston and rings, cylinder liner, valves and guides, turbocharger inlet and outlet, charge air cooler inlet and outlet, intake and exhaust manifolds, and any other air and exhaust gas passages of the engine. Other engine components and engine areas affected by low load operation should also be subject to this requirement for visual documentation.

d. Temperature and pressure readings necessary for complete evaluation of the engine performance including ambient room temperature, barometric pressure, and relative humidity. Fuel consumption in kilograms per hour (kg/hr) (pounds per hour [lb/hr]) should be entered for all runs of a duration for engine conditions to become stabilized. Additions of lubricating oil and oil changes. Readings of instruments at intervals not exceeding 1 hour; should not be required when a phase of a test is of such short duration that operating temperatures cannot be stabilized.

e. Fuel consumption. Fuel consumption data submitted in tabular form, and in the form of curves as follows:

(1) Specific fuel consumption in lb/hr, exhaust gas temperature, injection pump rack position (if practical), and air box pressure versus BHP. Data should include all speeds and loads specified in the test procedure.

(2) Fuel consumption at no-load in lb/hr versus speed.

(3) With BHP as the ordinate and RPM as the abscissa, draw lines of constant BMEP, and super impose curves of constant fuel consumption.

(4) To determine the control of the load limiter (when applicable) on engine output, a curve of maximum BHP or BMEP versus RPM for the operating range of the engine should be furnished.

f. Spectrometric analysis. The following information/data should be requested:

(1) List of oil wetted wearing parts and corresponding wear metal elements.

(2) Recommended limits in ppm of the wear elements/metals listed in (a). It is the intent to use these values as a basis to segregate normal from abnormal operating machinery.

(3) Tabulation should include such related and pertinent data as chemical and physical characteristics of the lubricating oil used during the test, initial quantity of oil in the system, hours of operation from start of test at which oil was added/changed, and applicable quantity of oil added/changed.

g. Power correction and performance curves. For generator sets, the value corresponding to each specified rated load or torque percentage will be herein referred to as the target load. Engine operation at each specified target load will be achieved by adjusting the observed load until the corrected value of the observed load is within ±1 percent of the specified target load. The 100 and 110 percent, full-load, power (BHP) must be greater than or equal to the proposed rating for the test engine. The average of the recorded power data values for the 100 and 110 percent power operating conditions must be greater than or equal to 100 and 110 percent of the specified rated power, respectively.

h. Fuel efficiency and thermal load results. The fuel efficiency and thermal load results must be recorded. The plotting guideline is as follows:

(1) Both BHP vs. RPM and brake torque vs. RPM curve.

(2) Constant Brake Specific Fuel Consumption (BSFC) “isometric map” plotted on the brake torque vs. RPM engine load profile curve. The propeller load curve (percent engine power vs. percent design hill speed) for ship design speed and corresponding engine/propeller speed (if applicable) must also be shown.

(3) Combined exhaust turbine inlet temperature “isometric map” plotted on the brake torque vs. RPM load curve.

(4) Intake air depression, charge air pressure, and exhaust back pressure plotted on the brake torque vs. RPM load curve.

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6.3.7 Shutdown report. For any shutdown, recommend that the Technical Authority be notified in writing as soon as the shutdown occurs. Additionally, a report describing the investigation and root cause of the shutdown should be provided by the manufacturer to the Technical Authority. This engineering report with supporting documentation, analyses, and rationale detailing the cause, corrective action, and impact on the test completion criteria should be submitted to the Technical Authority for evaluation to determine if additional testing is warranted. The Technical Authority will determine if the shutdown will be considered a forced shutdown or permissible shutdown. If the failure is attributable to the design of the component, then the shutdown should be classified as a forced shutdown. Failures attributable to manufacturing or quality defects may require recommencement of the entire endurance test depending upon the results of an investigation of the manufacturer's quality system. Endurance testing may continue at the manufacturer's risk while the failure investigation is ongoing.

6.4 Drawings.

6.4.1 Installation drawings. Installation drawings must include the following:

a. Overall dimensions for the complete assembled unit

b. Foundation bolting dimensions, fit requirements, material specifications, and installation procedure

c. Wet and dry weights, including detached accessories

d. Center-of-gravity in at least two planes

e. Both minimum and desirable clearance for removal of parts such as cylinder heads, pistons, liners, cooler tube bundles, camshafts, pumps, and blowers

f. Bill of material describing make, model, and quantity of equipment and accessories being furnished

g. Capacity of blowers and pumps at rated engine speed and volume of exhaust gases at rated speed and load specifying pressure and temperatures for which data apply

h. Show BHP, BMEP, and RPM

i. Location, type, and size of pipe connections to engine and accessories

j. For engines having detached accessories:

(1) Diagrammatic sketches of engine piping systems (e.g., sea and freshwater, lube oil, fuel oil, and air starting) indicating pipe sizes, type of fittings, and relative location of coolers, filters, valves, strainers, temperature/pressure indicators, thermometers, and gages. Manufacturer furnished piping must be shown by solid lines and installing activity furnished piping by broken lines.

k. Heat balance chart (unless shown in the operator's manual)

l. Rotation at power output

m. Gear reduction

n. Engine fluid capacities (including gear if furnished)

o. Air consumption

p. Alignment procedure

6.4.2 Accessory drawings. Accessory drawings must include the following data.

a. A cross-section drawing with bill of material must be sufficient for the following:

(1) Regulating and overspeed governors

(2) Cold start device

(3) Lube oil filter

(4) Thermostats or temperature regulators

(5) Starting system components

(6) Air intake filter/silencer

(7) Pneumatic control devices

(8) Fuel oil filter and strainer

(9) Mufflers

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(10) Coolers

(11) Water filter

b. An outline drawing will suffice to complete drawing requirements for the following accessories:

(1) Tachometer

(2) Tachometer generators

(3) Pressure gages

(4) Thermometers

(5) Thermocouples

(6) Flexible exhaust hose

(7) Expansion tanks

(8) Valves (relief safety, throttling, regulating, reducing, and similar types)

c. When resiliently mounted equipment is provided (see 3.7.13), a drawing must include the following information:

(1) Speed range (for propulsion machinery and direct current [DC] generators) or synchronous speed (for alternating current [AC] generators of the unit).

(2) Total weight of the mounted assembly in the operating condition. The weight must include weight of subbase, service fluids, piping filters, and all other attached accessories.

(3) Location of the unit center-of-gravity in at least two planes.

(4) The moments of inertia and products of inertia of the mounted unit about three mutually perpendicular axis with the origin at the unit center-of-gravity and the orientation of the axis indicated with respect to the equipment and ship.

(5) The six natural frequencies (in Hz) of the unit.

(6) The type mounting used in performance of the calculation. For mounting other than approved Navy type, information relative to mount natural frequency, static load deflection, and transmissibility is required.

(7) List of assumptions made in calculating natural frequencies.

6.5 Manuals. If technical manuals are required, they should comply with MIL-DTL-24784. The technical manuals should be acquired under separate contract line item in the contract. Manuals must include the following:

a. Engine installation drawing.

b. Engine sectional drawings.

c. Functional description.

d. Operation.

e. For threaded holes, the dimensions, including thread data and length, and complete instructions covering the removal and replacement of all inserts used must be included in the engine instruction manual.

f. Table of normal clearances, diameters and thickness for new parts, and maximum allowable clearances and wear limits for all wearing parts.

g. Torque settings of important threaded hardware such as fasteners using locking devices or fasteners securing moving parts and fasteners at gas or fluid boundaries.

h. Linear vibration limits and locations on the engine, lube oil, and jacket water analyses along with trending limits, torque settings, and hydraulic pressures of important threaded hardware and those requiring shipboard level maintenance.

i. Periodic maintenance.

j. Corrective maintenance.

k. Troubleshooting.

l. For a complete propulsion or generator unit including one or more engines, couplings, generator, or reverse reduction gear, the manual must include complete and comprehensive alignment instructions for the entire plant.

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The instruction must cover the original alignment procedure with maximum limits for radial and axial deviation and information for alignment checks after installation. A machinery layout sketch showing locations of measurements and space for recording measurements must be provided.

m. Description of valve with details of function and design limits.

n. Parts list.

o. Copy of type test certification.

p. Installation instructions.

q. Maintenance in service instructions to include testing and renewal of any sealing arrangements.

r. Actions required after a crankcase explosion.

6.5.1 Validation. The manufacturer should be responsible to validate or have validated the information contained therein for all manuals supplied. In those cases where validation is performed at a manufacturer's plant, the manufacturer is responsible for the quality of the technical manual information and must provide verifiable evidence of the adequacy of the validation performed. Validation must be completed and all corrections made and validated prior to the presentation of the technical manual to the Technical Authority for verification. Personnel performing the validation must be on a technical level equivalent to the intended user of the technical manual. Technical manuals or any part of a technical manual that has been validated for any other Technical Authority requirement need not be re-validated. Validation must include the following:

a. All written information, engineering drawings, and artwork in the manual must be compared to the related physical equipment or system to ensure that they, in fact, delineate that equipment or system.

b. A demonstration of the manual on the performance of all instructions and procedures for the equipment supplied except for the following:

(1) Destructive testing or destructive disassembly is not required.

(2) Boring, grinding, or other shaping repair procedures are not required.

(3) Checking for accessibility may be determined by measurement, observation, and reference to drawings.

6.5.1.1 Validation page. A validation page must be included in each manual supplied, inserted immediately after the title page, including the following information:

a. Manufacturer's full identification (if other than prime, full identification of both must be indicated)

b. Contract number(s)

c. Chapters and sections that were validated and the date each was accomplished

d. Chapters and sections that were not validated

e. Name, signature, and authority of validating officer

6.5.1.2 Validation of engine dimensional clearances. The manufacturer should be responsible for validating the external systems (i.e., starting, cooling water, fuel, lubricating oil, air intake, exhaust, drive systems, and mechanical or electrical accessories) of each engine and providing minimum and desirable dimensional clearance requirement data to facilitate access to various parts of each when adjustments and repair are needed.

6.5.1.3 Validation of special tools dimensional clearances. The manufacturer must validate the correct operation of all special tools associated with each engine and provide minimum and desirable dimensional clearance data required for each special tool to optimize accessibility in performing the necessary maintenance and repair functions.

6.6 Definitions. For the purposes of this specification, the following definitions apply.

6.6.1 Load factor. Percentage of time that the engine may operate at the rated brake horsepower and speed.

6.6.2 Maintenance engineering analysis (MEA). An analysis of the engine depicting the location, weight, and dimensions of all major components that may be removed for maintenance. This analysis typically consists of various engine schematics that depict this information, along with showing the space required to remove the major parts.

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6.6.3 Metal edge. Stacked metal discs or helically wound flat metal wire or metal ribbon media. Wire cloth, screen sintered metal, or perforated plate media are not acceptable for simplex strainers.

6.6.4 Navy qualified engine. An engine for which a letter has been issued by the NAVSEA Technical Authority, where the performance standard is indicated in the letter and is equivalent to those standards specified in this defense specification (see 4.2.6.1).

6.6.5 Rotation. For engines procured under this specification, rotation is defined as being clockwise or counterclockwise as viewed facing the power output end of the engine.

6.6.6 Significant changes. Significant changes are modifications made to the engine components or system that affect engine performance, shaft dynamics, or cycle temperatures. Additionally, any engine component modified such that it is no longer interchangeable with the component it replaces is considered to have undergone a significant change.

6.6.7 Special tools. Those tools not listed in the GSA Global Supply Catalog.

6.6.8 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.7 Spare parts. Recommendations must be determined as to the spare parts, gaskets, and all other consumables except fuel and lubricants, which would be required to support each engine unit with all its components and equipment for a period of 90 days. This recommendation must assume that operation during the 90 days is at a constant 80 percent load and 100 percent speed condition. The materials list must include not only the items to be replaced because of wear or breakage, but also items that are consumed in the course of the examination adjustments (e.g., gaskets and filters). Recommended maintenance operations for the proper care of the equipment must also be considered.

6.7.1 Listing insurance items. List the insurance items that are recommended to be carried onboard the ship that provide an optimum “get home” capability during the above same 90-day operation (mission).

6.7.2 Criteria for determining recommended spare parts. The following criteria must be used in determining recommended onboard spare parts and related consumables:

a. Demand based items (items having a predicted usage of at least one unit in 90 days for all installations onboard a ship).

(1) Provide an effectiveness (filling of demands onboard) of 90 percent for a period of 90 days.

(2) Be predicated on combat consumption rates wherever such rates can be ascertained.

b. Insurance items (items that do not have a predicted usage onboard ship of at least one in 90 days).

(1) Only those insurance items essential to end item performance and vital to the support of the primary mission of a ship or unit or vital to the safety and welfare of personnel onboard ships must be provided.

(2) Insurance items must be included in minimum depth (either unity or minimum replacement unit to meet potential 90-day combat requirements).

6.7.3 Critical spare parts. Identify parts that are deemed as critical spare and spare parts. A part can be classified as critical when failure of the part results in one of the following conditions:

a. Severe damage or immediate shutdown

b. Reduced performance capabilities of the engine

c. The potential exists for reduced capabilities, failure of the engine, or the creation of an unacceptable safety condition

6.8 Bid data. Complete engine and accessory data must be furnished with bids in accordance with the list below. This information is of value to the shipbuilder for piping and structural design purposes. Shipbuilders must furnish the data below, covering the engine, as advance information to establish engine-to-hull records to the appropriate organization and location.

a. Manufacturer Name

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b. Model

c. Classification Approval

(1) Classification Authority

(2) Approval Date

(3) Current Marine Applications

d. Configuration Type

(1) Application (main propulsion diesel engine, standby service diesel generator, auxiliary, and emergency diesel generator)

(2) Number of cylinders

(3) Rotation (clockwise/counterclockwise)

(4) Bore and stroke

(5) Cycle (2 or 4)

(6) Arrangement (inline or vee)

e. Overall dimensions

(1) Length

(2) Width

(3) Weight

f. Weights

(1) Engine only (dry)

(2) Engine only (wet)

(3) Subbase

(4) Accessories

g. Continuous duty rating

(1) BHP/power

(2) RPM (idle and max rated speed)

(3) BMEP

(4) Specific fuel consumption (mass and volumetric)

(5) Volume of exhaust gas (rated load)

h. Exhaust emissions

(1) EPA tier rating and certification

(2) IMO rating

(3) Emissions (criteria pollutants) g/kW-hour

(4) Muffler (manufacturer name, type, and capacity or range)

(5) After-exhaust treatment (manufacturer name, type, weight, and fluid or power requirements)

i. Combustion air system

(1) Blower or turbocharger type (manufacturer name and quantity per engine)

(2) Intercooler or aftercooler (manufacturer name, type, material, and coolant [raw or jacket water])

j. Control system

(1) Governor (manufacturer name and type)

k. Crankcase scavenging

(1) Method

(2) Pressure

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(3) Crankcase explosion protection (manufacturer name and type)

l. Fuel oil system

(1) Approved fuels

(2) Restricted fuels

(3) Fuel service pump (manufacturer name, type, and capacity/pressure)

(4) Fuel injection system (manufacturer name, type, and pressure)

m. Freshwater (jacket water) system

(1) Capacity in liters (gallons)

(2) Water treatment (manufacturer name and type)

(3) Freshwater pump (manufacturer, type, engine drive/electric, quantity per engine, capacity including volumetric flow rate, and pressure)

(4) Freshwater cooler (manufacturer, type, material, capacity, flow rate, and pressure)

n. Lubricating system

(1) Wet sump

(2) Dry sump

(3) Lube oil capacity

(4) Recommended lube oil

(5) Recommended change interval

(6) Lube oil pressure pump (manufacturer, type, and capacity)

(7) Lube oil scavenging pump (type, manufacturer, and capacity)

(8) Lube oil cooler (manufacturer, type, material, capacity, flow rate, and pressure)

o. Sea water system

(1) Sea water pump (manufacturer, type, engine drive/electric, quantity per engine, capacity, flow rate, and pressure)

p. Starting system

(1) Type (air, electric, or hydraulic)

(2) Manufacturer

(3) Capacity

(4) Starter type (crankshaft motor or in-cylinder)

(5) Number of starters

(6) Pressure/voltage

(7) Storage required (cubic feet and amp hours)

(8) Starting aid keep-warm system (manufacturer and type)

6.9 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 only applies to the lowest repairable units.

6.10 Subject term (key word) listing.

Auxiliary service set
Generator set
Governor
Jacket water
Low load
Overspeed
Turbocharger

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6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CC

Preparing activity:
Navy – SH
(Project 2815-2025-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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